

Dendritic morphology as a dictionary for local credit assignment

Houman Safaai^{1,*}, Maceo Richards¹, and Bernardo L. Sabatini^{1,2,*}

¹Kempner Institute for the Study of Natural and Artificial Intelligence at Harvard University, Cambridge, MA, USA

²Department of Neurobiology, Howard Hughes Medical Institute, Harvard Medical School, Boston, MA, USA

*Correspondence: houman_safaai@harvard.edu; bernardo_sabatini@hms.harvard.edu

Abstract

Dendritic trees compartmentalize synaptic inputs, but which spatial distinctions a learning signal must preserve is unclear. We describe delivered credit as combinations of spatial profiles constrained by morphology. In image classifiers with one dendritic neuronal layer, preserving neuronal identity accounts for most of the feedback benefit. Contextual conflict favors branch selection, while hierarchical distractors favor ancestry grouping at a bandwidth predicted by the task. Targets with matched input-sensitivity spectra develop different learned credit spectra and tolerance of fixed profiles; a local inhibitory gate also supports learning of opposing branch tuning in a conductance tree. Serial compartments help when nuisance-factor information is distributed across modules. Reconstructed arbors compress modeled passive perturbation fields through ancestry profiles, and focal shunts regulate their gains with selectivity set by electrotonic state. Measured visual responses provide no evidence of preferential ancestry alignment in the sampled cells. The framework links task structure to useful credit distinctions while separating available delivery patterns from the mechanisms that generate their coefficients.

Introduction

Learning must link a synapse’s local activity to its contribution to a circuit outcome. Cable filtering, nonlinear integration and spatially organized inhibition make locations within a dendritic tree functionally distinct [1–7]. These differences expand computation. We ask when branching also helps distribute useful learning signals.

Credit assignment asks how changing a synapse would affect task error. Backpropagation (BP) answers this by calculating the loss gradient—the sensitivity of task error to each parameter [8]; related sensitivity calculations apply to recurrent and steady-state networks [9, 10]. For a point neuron, each weight gradient combines a local sensitivity, called eligibility, with a neuron-specific learning signal. Hebbian plasticity captures the association between local pre- and postsynaptic variables [11], while three-factor rules combine that eligibility with an outcome-dependent signal [12]. We call a rule local when a synapse combines its own eligibility with an available learning signal, whether or not that signal is computed locally.

Biologically motivated learning theories differ in how they obtain such signals. Feedback alignment communicates errors through separate pathways [13, 14]; equilibrium, predictive and prospective methods infer target states or errors from dynamics [15–17]; eligibility propagation combines temporal traces with neuron-specific signals [18]. One number shared across neurons carries less task information than separately assigned neuronal signals, although each synapse still has its own eligibility [19–21]. Recent causal evidence for distinct instructive signals across neuronal populations motivates asking what additional distinctions are useful within a neuron’s arbor [22].

A *route dictionary* describes the spatial profiles available for delivering learning signals. A route identifies the locations reached; its profile specifies their relative amplitudes, and a trial-dependent coefficient scales that profile. Each dictionary column stores one profile. A shared profile broadcasts throughout an arbor; combining several profiles can distinguish subtrees. We use *basis* for linearly independent profiles and *dictionary* when profiles may be redundant. Morphology constrains the available routes; the task and teaching mechanism determine useful coefficients.

Conductance changes the gains of anatomical delivery routes. Excitatory and inhibitory contacts open nonnegative conductances whose reversal potentials determine current; inhibition can lower input resistance and attenuate other inputs even when its own current is small [23–27]. Rescaling a delivery channel preserves its spatial profile, whereas altered cable transport can reshape how local voltage affects somatic output and task error.

Dendritic models already connect nonlinear compartments and structural plasticity to memory capacity [28], and branch-specific plasticity can organize nonlinear feature combinations [29]. Successive computations in artificial trees and detailed cortical models also establish substantial forward capacity [30, 31]. Dendritic learning theories supply mechanisms for somatic prediction, segregated teaching, interneuron-mediated errors, bursts and prospective dynamics [32–38]; cable-learning models optimize reconstructed neurons using synaptic sensitivities [39]. We ask which spatial distinctions these mechanisms must preserve within an arbor.

The conductance-tree gradient factorization and LocalCA rule were introduced in our predecessor preprint [40]. This Article extends those foundations with controlled task-to-credit comparisons, matched-spectrum and conductance learning experiments, and anatomical transport and response tests.

We test a progression from neuronal identity to branch selection, ancestry grouping and example-dependent credit across interacting branches. Controlled learning tasks establish useful distinctions; forward-compatibility tests separate learning gaps from representational limits. Reconstructed arbors then test anatomical delivery capacity and conductance-dependent gains, while measured responses bound the biological interpretation.

Results

Exact gradients separate local sensitivity from downstream credit

For a point neuron $y_u = f_u(\sum_i w_i x_i + b_u)$, the chain rule gives $\partial \mathcal{L} / \partial w_i = x_i f'_u \delta_u$, where x_i and w_i are synaptic input and weight, b_u is bias, and $\delta_u = \partial \mathcal{L} / \partial y_u$ is the neuronal error. The derivative f'_u is evaluated at the current input sum. Local eligibility $x_i f'_u$ and task-derived credit therefore supply different information (Fig. 1A).

The rooted-tree model has soma $n = 0$, compartment voltage V_n , and nonnegative synaptic conductances g_i . A contact with reversal potential E_i^{rev} and nonnegative activity x_i contributes current $g_i x_i (E_i^{\text{rev}} - V_n)$. In the directed artificial trees, child outputs drive their parents and a terminal-to-soma pass computes the steady state. Total input resistance R_n^{tot} is the reciprocal of the summed leak, excitatory, inhibitory and child-coupling conductances (Methods). Differentiating the current balance yields

$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n)}_{\text{local eligibility}} \underbrace{\delta_{n,u}^V}_{\partial \mathcal{L} / \partial V_n}. \quad (1)$$

Excitatory (E) and inhibitory (I) contacts share this factorization (Fig. 1A, filled blue and filled carmine contacts on the arbor). Eligibility tells how conductance changes local voltage; compartment credit tells how that voltage changes task loss.

Voltage error denotes a loss derivative. Each compartment has one directed path to the soma, giving

$$\delta_{n,u}^V = \delta_{0,u}^V \tilde{\alpha}_n, \quad \tilde{\alpha}_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}. \quad (2)$$

Here $f'_i(V_i)$ is the child compartment’s output derivative, $g_{i \rightarrow k}^{\text{den}}$ is its coupling conductance to parent k , and $\delta_{0,u}^V = f'_0(V_0) \delta_u$ is the somatic voltage error. The path multiplier $\tilde{\alpha}_n$ converts one somatic error into a spatial field. The conductance-only component of this product is attenuated by additional conductance on a path’s ancestors; branch nonlinearities and forward voltages can change as well (Supplementary Section S1 and Fig. S1).

Reciprocal cables instead require a joint adjoint calculation of how local currents affect task loss (Methods). These mathematical sensitivities do not establish a biological error-transmission mechanism.

Local credit assignment (LocalCA) retains eligibility and supplies an approximate compartment signal, denoted by a hat (Fig. 1B shows a strict scalar, one signal per neuron and the exact path, each on a generic tree):

$$\widehat{\nabla}_{g_i} \mathcal{L} = x_i R_n^{\text{tot}} (E_i^{\text{rev}} - V_n) \widehat{\delta}_{n,u}^V, \quad (3)$$

$$\widehat{\delta}_u^V = A_u \mathbf{c}_u. \quad (4)$$

Columns of $A_u \in \mathbb{R}^{N_u \times K}$ are fixed spatial delivery profiles over N_u routed nonsomatic compartments; $\mathbf{c}_u$ contains their example-dependent coefficients. To distinguish coefficient generation from delivery gain, we can write $\mathbf{c}_u = \Gamma_u \mathbf{c}_u^{\text{source}}$, where diagonal Γ_u scales each profile and $\mathbf{c}_u^{\text{source}}$ is the supplied signal (Fig. 1A, gain dial). Fixed nonzero gains leave

the dictionary span and its oracle capture unchanged while altering conditioning and reliability-weighted learning (Supplementary Figs. S4–S5). Broadcast, subtree and compartment profiles give increasingly detailed spatial addresses (Fig. 1C). This is a delivery model, distinct from the forward cable equations and the mechanism estimating coefficients.

At $K = 1$, the all-ones column $A_u = \mathbf{1}$ broadcasts one coefficient through the arbor; at $K = N_u$, the identity matrix $A_u = I$ can reproduce the full nonsomatic field. Nested subtrees provide intermediate dictionaries. Profile count, rank (the number of independent profiles) and coefficient information are distinct resources. Exact path transport combines one somatic scalar with state-dependent local derivatives; it does not require an independent external error for every compartment.

Capture is the fraction of squared learning-signal magnitude that a dictionary can reproduce. For a nonzero field δ ,

$$\mathcal{C}(A; \delta) = \frac{\|P_A \delta\|^2}{\|\delta\|^2}, \quad P_A = AA^\dagger.$$

Here P_A gives the least-squares reconstruction, $A^\dagger$ is the Moore–Penrose pseudoinverse and $\|\cdot\|$ the Euclidean norm. Capture measures field energy rather than decision accuracy. Averaging these per-field ratios weights fields equally; $\text{tr}(P_A C) / \text{tr}(C)$, with $C = \mathbb{E}[\delta \delta^\top]$, is the fraction of total ensemble energy captured, weighting fields by their energy. The uncentered second-moment matrix C has trace $\text{tr}(C)$, the sum of its diagonal entries. These capture measures can differ, and both assume optimal projection coefficients rather than an implemented encoder.

The field coordinate also matters. In nonlinear image models, we route errors of compartment outputs and retain the local output derivative in eligibility; voltage-error capture is reported separately. Identity-output conductance trees use voltage errors directly. Supplementary Table S2 compares the coordinates, coefficient sources and state dependence. Learning experiments test whether this available capacity improves performance.

Projection can also filter update noise outside a dictionary’s span. Retaining most of the useful gradient while admitting less noise can improve a one-step guarantee (Supplementary Fig. S3G). If the span contains the entire gradient, its mean is preserved. A one-step smoothness bound formalizes this balance between retained descent and update variance (Methods; Supplementary Fig. S3G). Its initialization utility did not select the best final morphology: in a separate 12,800-fit prospective experiment, it lost to a fixed maximum-budget tree in all twenty seed averages in both selection arms (Supplementary Fig. S34). We therefore use the bound to interpret local filtering, not to select a final tree.

One signal per neuron captures most of the image-task learning benefit

Image classification first tests whether within-arbor resolution helps when neuronal errors are available. All image networks used one dendritic layer and a trainable readout, which supplied exact neuronal output errors before the primary feedback interventions. The MNIST comparison used 128 directed $[3, 3]$ trees, each with three proximal and nine terminal compartments, receiving flattened images. Shunting and raw-additive architectures shared the graph, initial contact construction, nonlinearities and training protocol, with architecture-specific initialization. The additive control removed reversal-potential driving forces and the conductance denominator, retaining signed current summation without shunting.

Six rules separated neuronal identity, spatial resolution and learning in the dendritic core (Fig. 1B–D): a strict scalar, per-neuron feedback (neuron-specific), a projected $K = 1$ rule, projected $K = 3$ subtree profiles, the exact path and a decoder-only reference. The strict scalar shared one signal across neurons; per-neuron feedback broadcast each neuron’s own output error through its arbor. The projected $K = 1$ rule averaged the exact activation errors over all twelve nonsomatic compartments, while the projected $K = 3$ profiles averaged them within each proximal compartment and its three children. Per-neuron feedback uses the same all-ones profile as the projected $K = 1$ rule and differs only in the coefficient source (Fig. 1C). Both used the same exact field and preserved its mean at identical parameters, while allowing different norms. The exact path retained every compartment error. A decoder-only reference trained the readout while freezing the core. Three development seeds selected each rule’s rate from three common multipliers; ten fresh paired seeds per architecture evaluated selected and original common rates over 180 epochs (Methods; Supplementary Fig. S7A,B).

At selected rates, per-neuron feedback exceeded the strict scalar by 9.23 percentage points in shunting trees and 7.46 points in additive trees, reaching 97.12% and 97.11% accuracy (Fig. 1D,E). Decoder-only learning reached 80.96% and 89.57%, respectively (Fig. 1D). Increasing the projected dictionary from one to three profiles changed accuracy by only 0.049 points in shunting trees (95% paired interval, -0.048 to 0.141) and 0.036 points in additive trees (-0.056 to

0.122; point estimates in Fig. 1D, note; intervals in Source Data). The common-rate comparison gave similarly small differences (Supplementary Fig. S7C). The exact path added 0.010 points over per-neuron feedback in shunting trees (5 of 10 seeds positive) and 0.182 points in additive trees (0.114–0.252; positive in all ten seeds; Fig. 1F); against the projected $K = 3$ rule it was 0.09 and 0.04 points lower (Fig. 1D, note). Both the exact path and one projected profile reached about 97.3% in additive trees. Most learning benefit therefore came from preserving neuronal coordinates; the smaller additive improvement was attainable without separating subtrees.

The richer dictionary nevertheless captured more of the trained activation-error field in the same cohort. Moving from one to three profiles raised mean capture from 0.622 to 0.655 in shunting trees and from 0.764 to 0.859 in additive trees, and the twelve-profile identity dictionary reproduces the field exactly by construction (Fig. 1G). Additive capture nevertheless fell over training at both budgets (0.851 to 0.764 at $K = 1$; 0.893 to 0.859 at $K = 3$), and the two shunting values at initialization were indistinguishable (0.5322 and 0.5323; open markers in Fig. 1G). These ratios average over example–neuron fields and use the coordinates delivered during learning. Thus improved reconstruction did not yield a comparable accuracy benefit. Earlier voltage-credit diagnostics and recipient-permutation controls support spatial field structure and the importance of neuronal identity, respectively (Supplementary Section S2 and Figs. S8A–C).

With fixed random soma-level feedback (DFA), per-neuron feedback improved over the strict scalar by 11.03 points in shunting trees and 10.80 points in additive trees (fifteen paired seeds; Fig. 1E; Supplementary Fig. S6D), while exact paths added at most 0.09 points (Fig. 1F; Supplementary Fig. S6E). This control changes weight transport but retains the single dendritic layer and the same output-error dimensionality. Fashion-MNIST also favored per-neuron feedback over the matched-width scalar fallback, by 4.46 and 3.78 points, with exact paths within 0.14 points of it (ten paired seeds; Fig. 1E,F; Supplementary Fig. S6C). On flattened CIFAR-10, exact paths reduced accuracy relative to per-neuron feedback by 0.86 points (0.55–1.17), while remaining equivalent to matched backpropagation within the prespecified one-point margin (−0.115 points, −0.457 to 0.226; two one-sided tests $P = 1.6 \times 10^{-5}$; twenty paired seeds; Fig. 1F, diamond in the band; Supplementary Fig. S6A,B). These four cohorts differ in feedback source, scalar construction and tree (Methods).

Thus finer delivery need not improve learning even when the error field has spatial structure. Further depth and task-generator controls are in Supplementary Figs. S6 and S9A–D, which cover input coverage, depth and the historical noise generators; the serial-depth half is Supplementary Fig. S9E,F. We next introduce tasks that make specific spatial distinctions useful.

Branch-selective information prevents updates from inactive distractors

Branch selection is useful when otherwise valid local activity comes from irrelevant inputs. We tested this with one logistic output and $B \in \{2, 4, 8\}$ parallel branch weight vectors and no internal forward compartments or conductance nonlinearity (Fig. 2A). Context selected the branch determining the output, while every branch received an image and formed an ungated local eligibility.

Inputs were average-pooled, standardized Fashion-MNIST images from classes 0 and 6. The selected image determined the binary target. At conflict dose $\chi = 0$, all branches received that image; increasing χ independently replaced nonselected inputs with opposite-class images. For example i , let b_i^* be the selected branch, $\mathbf{x}_{i,b}$ its input vector and $y_i \in \{0, 1\}$ the target. The branch weights $\mathbf{w}_b$ give the pre-probability score (logit) and output error

$$z_i = \mathbf{w}_{b_i^*}^\top \mathbf{x}_{i,b_i^*}, \quad \delta_i = \sigma(z_i) - y_i. \quad (5)$$

Here $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function. Unselected branches have zero exact derivative despite nonzero eligibility. Conflict therefore challenges a rule that ignores selection by exposing it to irrelevant, opposite-class eligibilities.

For N examples, the branch-specific and neuron-shared gradient estimates were

$$\mathbf{d}_b^{\text{selected}} = N^{-1} \sum_i \delta_i \mathbf{1}[b_i^* = b] \mathbf{x}_{i,b}, \quad \mathbf{d}_b^{\text{shared}} = N^{-1} \sum_i \delta_i B^{-1} \mathbf{x}_{i,b}. \quad (6)$$

The indicator $\mathbf{1}[\cdot]$ is one when its condition holds, zero otherwise; $1/B$ matches scale without supplying branch information. A feedback selector and a branch-local eligibility gate give the same update (Fig. 2B), testing access to selection rather than a need for independently computed external errors. We call these the branch-specific and neuron-shared rules. A deranged route instead delivers the same signal to the neighbouring branch at matched rank and sparsity (Fig. 2B, bottom row).

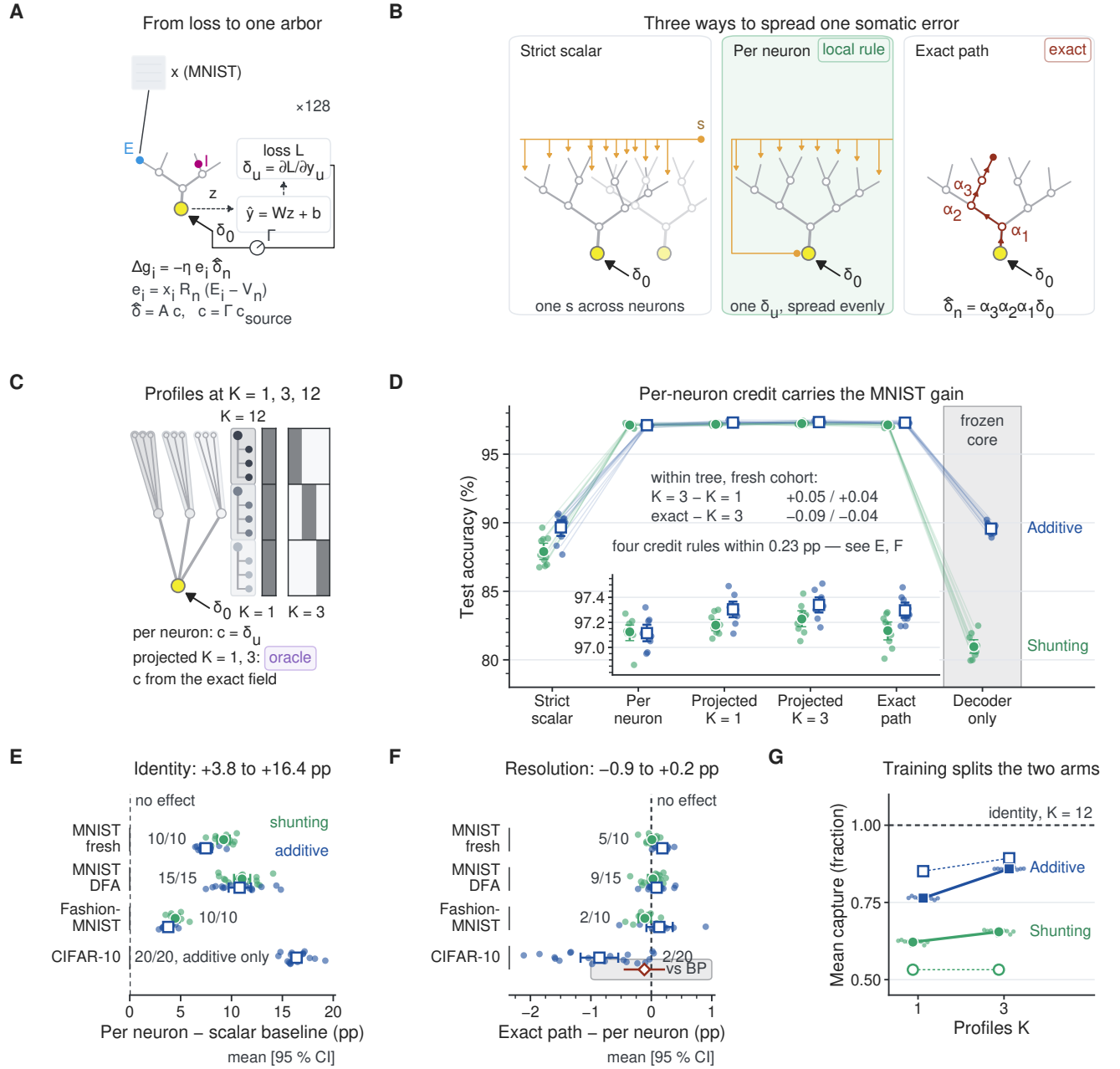


Figure 1: Neuronal identity, not finer spatial address, carries the image-learning benefit. **A**, MNIST input (leader) reaches an excitatory contact (blue) on one of 128 trees (the $\times 128$ tag); an inhibitory contact (carmine) is also drawn. The arbor is schematic, not C's [3, 3] morphology. Readout $\hat{y}$ and loss $\mathcal{L}$ return δ_u into the soma as δ_0 (ink arrow) through the diagonal gain Γ (dial), $c = \Gamma c_{\text{source}}$; no data. **B**, Three deliveries of one somatic error on a generic depth-three tree: a strict scalar s shared with a ghosted neighbour, one signal per neuron, and the exact path $\hat{\delta}_n = \alpha_3 \alpha_2 \alpha_1 \delta_0$; schematic, no data. **C**, Dictionaries over the twelve nonsomatic sites: the tinted strip is the $K = 12$ identity, beside $K = 1$ all-ones and $K = 3$ proximal subtrees; columns are indicator supports averaged by the projected rules (oracle badge); schematic, no data. **D**, Test accuracy for six rules at selected rates; circles shunting, squares additive, dots seeds paired by hairlines; band, decoder only (frozen core); note, the fresh-cohort within-tree contrasts; means with 95% seed-bootstrap intervals over ten fresh paired seeds per architecture, smaller than the symbol for the credit rules. Inset, those four credit-rule columns at the same x positions and marks, rescaled until their intervals and ten seeds resolve. **E**, Per neuron minus each cohort's scalar baseline (the strict scalar, except the matched-width fallback for Fashion-MNIST), shunting on the row and additive open below it, each row tagged with shunting positive seeds out of the total ($n = 10, 15, 10$ and 20 seeds; CIFAR-10 is additive only); means with paired 95% seed-bootstrap intervals in percentage points, several narrower than the symbol and hidden by it. **F**, Exact path minus per neuron in the same rows, marks and interval type; in the CIFAR-10 row the open red-brown diamond and interval are exact path minus backpropagation, inside the band of the prespecified ± 1 percentage-point margin. **G**, Mean activation-error capture of D's trained exact-path fields by C's dictionaries at $K = 1, 3$; filled, trained, with its ten seeds; open, initialization ($n = 10$, seeds not drawn); reference line at 1, the $K = 12$ identity, which reproduces the field. All 95% seed-bootstrap intervals are at most 0.007 capture units and are not drawn. pp, percentage points. Source Data: source_data/curated_publication/figure_01_plotted.csv.

With centered, symmetric class evidence, the shared-mode prediction crosses zero at $\chi_c = B/[2(B-1)]$: 1, 2/3 and 4/7 for two, four and eight branches (Fig. 2C; Methods). The measured initial signed utility of the neuron-shared update tracked this line to within 0.018 at all twenty-four sampled cells, although its bootstrap interval excluded the prediction in eleven of them (Fig. 2C). This is a statement about initialization: the trained chance crossing is a separate endpoint, displaced from the boundary because update alignment is lost during learning (Methods; Supplementary Fig. S10B–D). It nonetheless followed the predicted order $B = 8 < 4 < 2$ in 20 of 20 seeds, and five two-branch seeds never reached chance within the sweep (Fig. 2D). Selection also protects what has already been learned: in a separate two-stream interference task, neuron-shared credit erased a previously acquired context, losing 55.7 accuracy points [53.8, 57.6], while branch-specific subtree routes lost none in all ten seeds (Fig. 2E; Supplementary Fig. S10A).

At zero conflict, the branch-specific and neuron-shared rules achieved similar accuracy. Their difference increased with conflict, reaching 34.9–58.2 percentage points at full conflict across the tested branch counts (20 paired seeds; Fig. 2F–H). In every seed, the advantage of correct routing increased with conflict. A deranged route sent every signal to the wrong branch at matched rank and sparsity. It was harmful even at zero conflict, reducing accuracy by 12–25 points relative to branch-specific selection, whereas the neuron-shared rule was indistinguishable from branch-specific selection there (Fig. 2F–H, $\chi = 0$); at high conflict the deranged rule settled below chance, near 22% correct at every branch count, as did the neuron-shared rule at $B = 4$ and $B = 8$, while the two-branch neuron-shared rule reached 45.1%. BP and the gated point matched branch-specific selection exactly in every seed, as required by their algebraic equivalence (Fig. 2B; Source Data; Supplementary Section S3). Branch-count comparisons do not hold capacity fixed.

The neuron-shared rule reached chance at lower conflict as branch count increased, consistent with the predicted shift (Fig. 2D). Credit reaching inactive branches can therefore harm learning, and credit reaching the wrong branch harms it at every dose.

Ancestry provides a modest advantage at an intermediate feedback budget

Between shared and branch-resolved feedback, nested subtree routes offer an intermediate dictionary. The task contained eight independent, eight-dimensional terminal weight blocks at the leaves of a balanced binary feedback tree. Only the context-selected block determined the logit. Internal tree nodes defined feedback supports and did not perform serial forward computation (Fig. 3A).

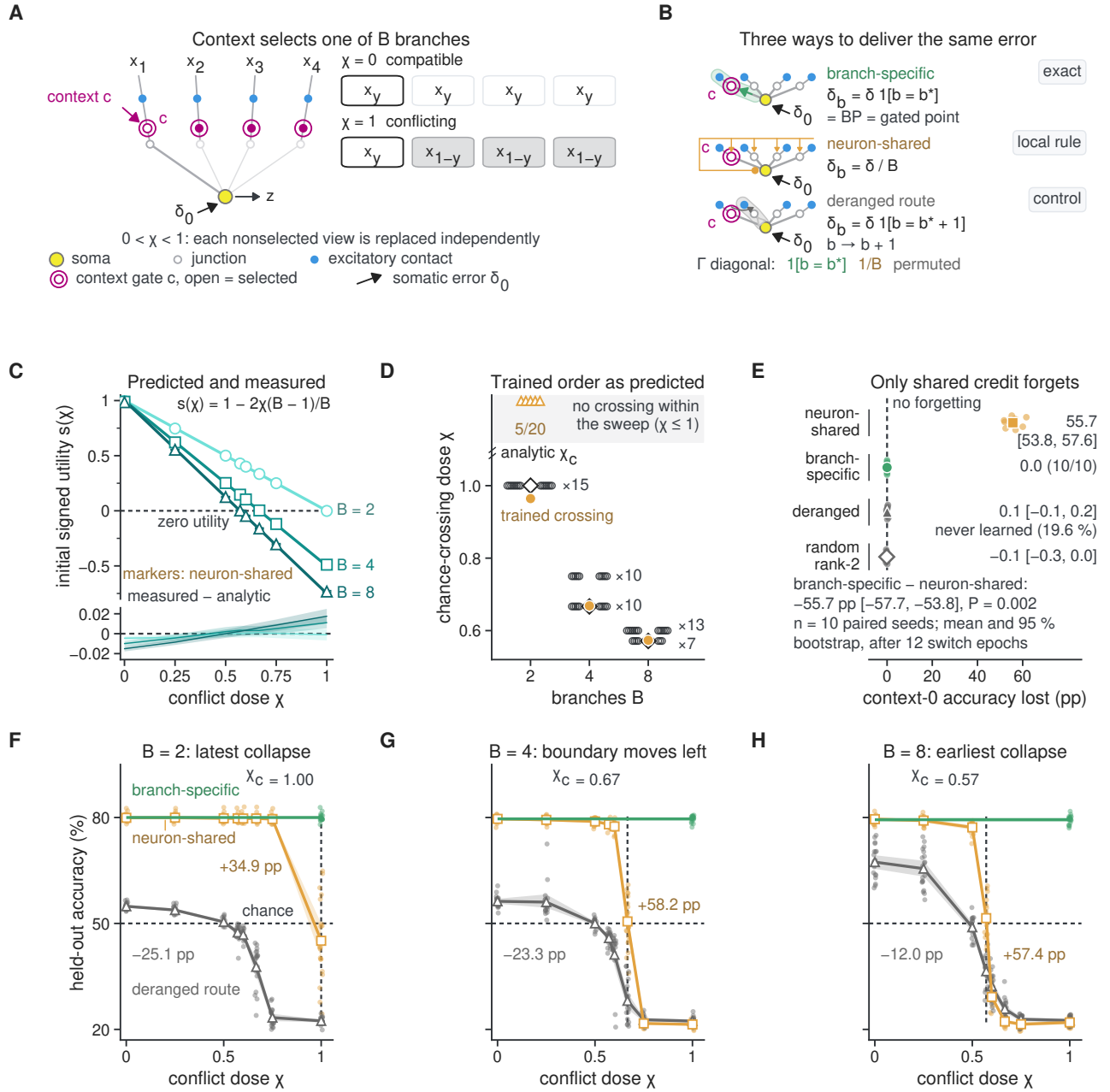
Distractors opposed the selected stream’s class signal increasingly with distance through the feedback hierarchy, deliberately favoring grouping that matches the task-to-route assignment. At each budget $K \in \{1, 2, 4, 8\}$, the routing field had K independent patterns and unit-length rows. Ancestry routes grouped the terminals into the whole tree, halves, sibling pairs or individual terminals (Fig. 3B). Permuting task streams across leaves changed the task-to-route assignment without changing the streams or forward parameter budget. Here bandwidth counts independent feedback patterns, not physical stages or independent scalar losses; other single-pattern controls need not be uniform. The complete generator and update are given in Methods.

The generator predicts the budget at which ancestry can help. The selected stream has class coefficient one; its sibling, same-half nonsiblings and other-half streams have coefficients -0.15 , -0.45 and -0.75 . The unnormalized sum of the class signal within the selected ancestry group gives -3.05 , -0.05 , 0.85 and 1 at $K = 1, 2, 4, 8$, respectively. Coarse pooling reverses or nearly cancels the signal; sibling pairs first preserve a positive intermediate-group signal, and full resolution removes grouping differences. The implemented unit-row normalization divides these sums by the square root of group size, preserving their signs. This budget follows from the design, not an independently discovered optimum (Fig. 3B, card footers).

Controls preserved rank while changing recipient assignment, support geometry or sparsity: cyclic derangement, depth-interleaving, random-sparse and dense random fields, whose delivered support of the cued stream’s credit is shown in Fig. 3B.

Increasing bandwidth improved correct routing, while derangement caused large losses (Fig. 3C). At $K = 1$ and $K = 2$ both fell below chance, as the design predicts: pooling reverses the class signal within the cued group (Fig. 3B). Matched-bandwidth partitions had coincident field capture, so capture does not explain the ordering (Supplementary Fig. S11A). Degree- and depth-preserving rewiring removed 23.15 and 5.02 points at $K = 2$ and $K = 4$ but left fully shared and fully resolved ancestry routes exactly tied (Fig. 3D).

The ancestry-specific effect was much smaller. Ancestry performed worse than the best matched control at low budgets, exceeded it by 1.27 percentage points at $K = 4$, and tied at full resolution. The paired $K = 4$ interval was



0.59–1.95 points, with 15/20 positive differences; a paired Wilcoxon signed-rank test gave Holm-adjusted $P = 0.0101$ across four budgets. Ancestry also exceeded the dense rank-4, random-sparse and depth-interleaved controls by 2.31, 3.77 and 7.11 points, and the deranged control by 61.06 points, so the small best-of-four contrast reflects a maximum taken over controls rather than a weak effect (Fig. 3E).

A separate twenty-seed cohort tested how supplied context cues could set the coefficients (Fig. 3F). An encoder trained on 256 calibration cues with noise SD 0.5 and their four-route activation targets improved task accuracy over a frozen profile by 7.16 points, but remained 54.47 points below oracle routing. Exploratory delivery of only its most probable route reduced this noisy-cue gap to 16.26 points. With noise-free cues the correct route was identified at every calibration size, and the leakage cost then depended on calibration alone: soft coefficients lost 60.90 points with sixteen cues but only 0.31 points with 256, whereas hard selection matched the oracle in all twenty seeds. Leakage into incompatible routes therefore mattered when calibration was small, not whenever the readout was soft. Hard selection retained a deficit under noisy cues, and delaying either readout across independent contexts removed its benefit (Supplementary Fig. S11B,C and Section S3).

Cross-branch interactions change the spatial credit that learning uses

Branch interactions can change credit’s sign and magnitude between examples. We isolated this effect using targets with identical input-sensitivity spectra on the same compatible tree.

Eight independent inputs x_i were equally likely to be -1 or $+1$. Each branch computed $u = a + bu_L + cu_R + du_Lu_R$ with four trainable coefficients. This multi-affine unit is affine in either child while the other is fixed. Pairwise targets summed four disjoint products, $f_M = \frac{1}{2} \sum_{(i,j) \in M} s_{ij}x_ix_j$; quartic targets combined complementary quartets, $f_A = \frac{1}{2}(s_A \prod_{i \in A} x_i + s_{\bar{A}} \prod_{i \notin A} x_i)$. Here M partitions inputs into pairs, A is one quartet, $\bar{A}$ its complement, and signs s were independently sampled. Within each seed, targets shared a compatible balanced tree, input assignment, initial weights and examples (Fig. 4A). Both have input-sensitivity second moment $\mathbb{E}[\nabla_x f \nabla_x f^T] = I_8/4$, giving identical strengths of independent sensitivity directions.

A scalar branch output cannot transmit several independently required input combinations to its parent. This communication constraint gives an exact compatibility criterion: at each subtree boundary, the matrix of interaction coefficients between inside and outside input products must have rank at most one after removing the constant inside row. The condition is necessary and sufficient across all boundaries; discarded singular values bound representation error (Methods; Supplementary Section S4).

Logical rules make this forward constraint concrete. All fifteen binary trees over four inputs can represent AND, OR and parity exactly, but mixed rules can require particular input groupings. The rule $(a \text{ AND } b) \text{ OR } (c \text{ AND } d)$ needs only two serial stages, whereas $a \text{ AND } [b \text{ OR } (c \text{ AND } d)]$ needs three in this scalar model, despite containing the same two AND gates and one OR gate. These representability statements are independent of training (Supplementary Figs. S14–S15). They identify possible morphologies before testing credit-dependent learning.

All credit rules used the same exact local eligibilities and scalar output error. We compared exact paths with unit broadcast and calibrated broadcast, whose six-site profile is calibrated from each student’s mean path derivatives over 256 unlabeled initialization examples (Fig. 4B). Three development seeds selected learning rates with equal budgets; twenty fresh paired seeds tested the frozen choices. A separate nested target summed products over prefixes of two, four, six and eight inputs on its own compatible tree, which requires depth four and does not share the $I_8/4$ input spectrum (Supplementary Fig. S12F).

Calibrated broadcast learned pairwise targets but remained inaccurate on quartic and nested targets (Fig. 4C–E). At the primary checkpoint (1,024 updates), mean normalized mean squared errors (NMSE) for exact path and calibrated broadcast were 0.0317 and 0.0236 on pairwise targets, 0.1516 and 0.9813 on quartic targets, and 0.0230 and 0.7690 on nested targets. NMSE divides squared prediction error by clean target variance. The predefined quartic-minus-pairwise difference in calibrated-broadcast-minus-exact gaps was 0.838 (95% paired interval, 0.719–0.931), positive in all twenty seeds; a common learning rate gave 0.843 (0.727–0.935). Exact quartic learning was heterogeneous at this checkpoint: seventeen fits were near the noise floor and three remained stalled, and those three resolve one at a time between 1,024 and 4,096 updates (Fig. 4D). The lower pairwise mean for calibrated broadcast likewise reflected one slow exact fit; exact path performed better in fifteen of twenty seeds (Supplementary Figs. S16–S17).

A balanced continuation of every algebraic trajectory to 16,384 updates tested whether these gaps persisted without



Figure 3: Ancestry routes help only where the tree's partition matches the task, and only when the coefficients are supplied. **A**, Eight streams enter terminals b_1 – b_8 of a balanced tree; context c selects b_3 , whose block sets the logit z , and the $K = 4$ capsule delivers δ_0 to b_3 and sibling b_4 . Junctions define feedback supports only; the tinted band gives each tier's coefficient by tree distance. **B**, Ancestry dictionaries A ($8 \times K$): K columns of $8/K$ terminals, the channel carrying b_3 's credit green and the others grey; $K = 8$ is the identity, one column of eight unit channels labelled $A = I$. Raw class-signal sums are -3.05 , -0.05 , $+0.85$, $+1.00$ (row normalization preserves the signs); beside them, b_3 's support under the four controls at $K = 4$ (random-sparse and dense rows, one draw). **C**, Held-out accuracy across budgets for the ancestry route, the deranged route and the best matched control (the per-seed maximum over the four controls, a ceiling); 20 paired seeds, means with 95% seed-bootstrap intervals, smaller than the markers where no bar is visible. **D**, Task-matched versus degree- and depth-matched rewired tree under ancestry feedback: +23.15 [21.03, 25.32] and +5.02 [4.19, 5.94] points at $K = 2, 4$, exact ties at $K = 1, 8$; 20 paired seeds. Bracketed intervals are paired 95% seed-bootstrap intervals, reported not drawn; the drawn bars are per-condition and smaller than the markers. **E**, Ancestry minus each control at $K = 4$; the top row is the per-seed best of the four controls, the three below it plus the off-scale deranged control, so its seeds repeat theirs; Holm-adjusted P across four budgets (top row) or four controls; 20 paired seeds, means with 95% seed-bootstrap intervals; the fan is clipped at 12 points, the seed beyond it named on the panel. **F**, Coefficient source across the calibration $\times$ cue-noise grid at zero delay: soft solid with filled marks, hard dashed with open marks at three noise levels (covered at noise 0, where the hard series meet), against the oracle ceiling and frozen-profile floor. At 256 cues and noise 0.5, soft is -54.47 [-55.45 , -53.54] below oracle and $+7.16$ [6.26, 8.07] above the floor; 20 fresh seeds, 95% seed-bootstrap bands. Teal ramps in **A** and **F** are ordinal within this figure alone; **C**–**F** report epoch 80 and **A**, **B** are schematics. Source Data: [source_data/curated_publication/figure_03_plotted.csv](#).

retuning. All twenty selected-rate Adam exact quartic fits reached the noise floor by 4,096 updates, and the pairwise mean lag disappeared. At 16,384 updates, the quartic-minus-pairwise calibrated-broadcast deficit was 0.952 (95% paired interval, 0.883–1.012) at endpoint states and 0.790 (0.691–0.880) at validation-selected states, positive in all twenty seeds at all four budgets (Fig. 4F). Calibrated-broadcast arms contacted coefficient bounds; exact-path arms did not. These are the same previously observed seeds and bounded dynamics, with common-rate, per-seed and optimizer controls (selected-rate Adam trajectories in Supplementary Fig. S16A,C; Supplementary Figs. S16–S17 and Table S8).

The trained fields reveal a distinction missed by the input spectrum. Let $\mathbf{q}(x)$ contain the six nonsomatic derivatives of root output with respect to branch output; exact branch credit is scalar output error times $\mathbf{q}$. At the 1,024-step checkpoint, allowing an oracle amplitude per example, the best rank-one profile captured 99.7% of exact-trained path-field energy for pairwise targets, but 46.1% for quartic and 44.8% for nested targets. The separation persisted at 16,384 updates: capture was approximately 100%, 39.6% and 40.5%, respectively (Source Data). Reaching 95% of the field energy took one fitted direction for pairwise targets and about five for quartic and nested targets. Loss-error weighting, eligibility weighting, site-normalized spectra and replay of 600 earlier fits preserved the original separation (Supplementary Section S4). Uniform-profile capture did not (0.122, 0.169 and 0.176; Fig. 4G, leftmost category): a nearly one-dimensional field can have a signed, nonuniform direction. Positive fixed-profile scaling largely cancels in Adam, explaining calibrated/sign-only agreement; it does not explain the slower pairwise learning with changing exact paths.

The learned credit spectra help explain these input-matched targets' different tolerance of calibrated broadcast. The results favor example-dependent spatial credit without establishing that all six exact channels are necessary.

Additional controls delimit the result. Morphology also matters within a fixed rule: shuffling which input enters which leaf, at fixed tree shape and parameter count, raised exact-path NMSE by 0.575 (95% paired interval, 0.557–0.592; positive in all twenty seeds) pooled over the three task families (Fig. 4H and Supplementary Fig. S13C). Calibrated-broadcast Adam learned a compatible positive-conductance teacher accurately; broadcast learned aligned XOR-of-AND, whereas parity favored exact credit under the tested optimizer (Supplementary Figs. S14–S15 and S18), and the 3,600-fit family-by-optimizer-by-profile screen behind these comparisons is in Supplementary Fig. S13. Noisy-query interaction estimation also yielded useful trees, retained as supporting structure-design evidence in Supplementary Fig. S36. Interaction structure therefore motivates specific credit comparisons rather than a universal monotonic resolution law.

Context-dependent branch tuning makes spatial credit useful in conductance trees

Conductance trees provide a biophysical model for testing this distinction. All rules learned a seven-compartment teacher with monotonic branch responses: extended Adam training kept broadcast NMSE below 0.0006 despite a varying path field (Supplementary Fig. S19). A separate construction tested context-selected branches with aligned or opposing feature tuning (Fig. 5A).

Four terminal compartments fed two proximal compartments and a soma, using the directed E/I conductance balance (Methods). Both subtrees received the same positive feature pairs, $\exp(z_j)$ and $\exp(-z_j)$, for two independent latent inputs z_j . A binary context increased inhibitory conductance in one proximal compartment. Teacher excitation and inhibition produced increasing feature responses in both subtrees for the aligned task, but increasing responses in one and decreasing responses in the other for the opposed task. Teacher and student shared the circuit, ensuring representability. Students shared the teacher's nominal proximal and coupling gains, with independent Gaussian perturbations of SD 0.4 in log conductance; terminal gains started near one. All 24 conductances were plastic, and paired conditions shared initial weights and example streams.

At fixed weights, binary context generates at most two exact path profiles. Their gains are positive: context changes the relative weighting of the two distal subtrees, while proximal credit is context invariant. The nominal parent's conductance denominator changes by approximately eightfold. Thus branch selection is the relevant distinction, connecting this task to Fig. 2A,B.

To supply selection without oracle coefficients, a local rule delivered the somatic error to terminal parameters only when their parent was uninhibited, while retaining ungated error at proximal compartments and the soma. It used the supplied inhibitory context and exact local eligibilities, without evaluating exact paths or projecting an exact field. Two fixed distal profiles with oracle amplitudes and unit proximal credit supplied a separate capacity control; the four deliveries of the somatic error are drawn together in Fig. 5B. An additional proximal profile in the earlier three-profile control therefore addresses proximal updates, rather than another context distinction.

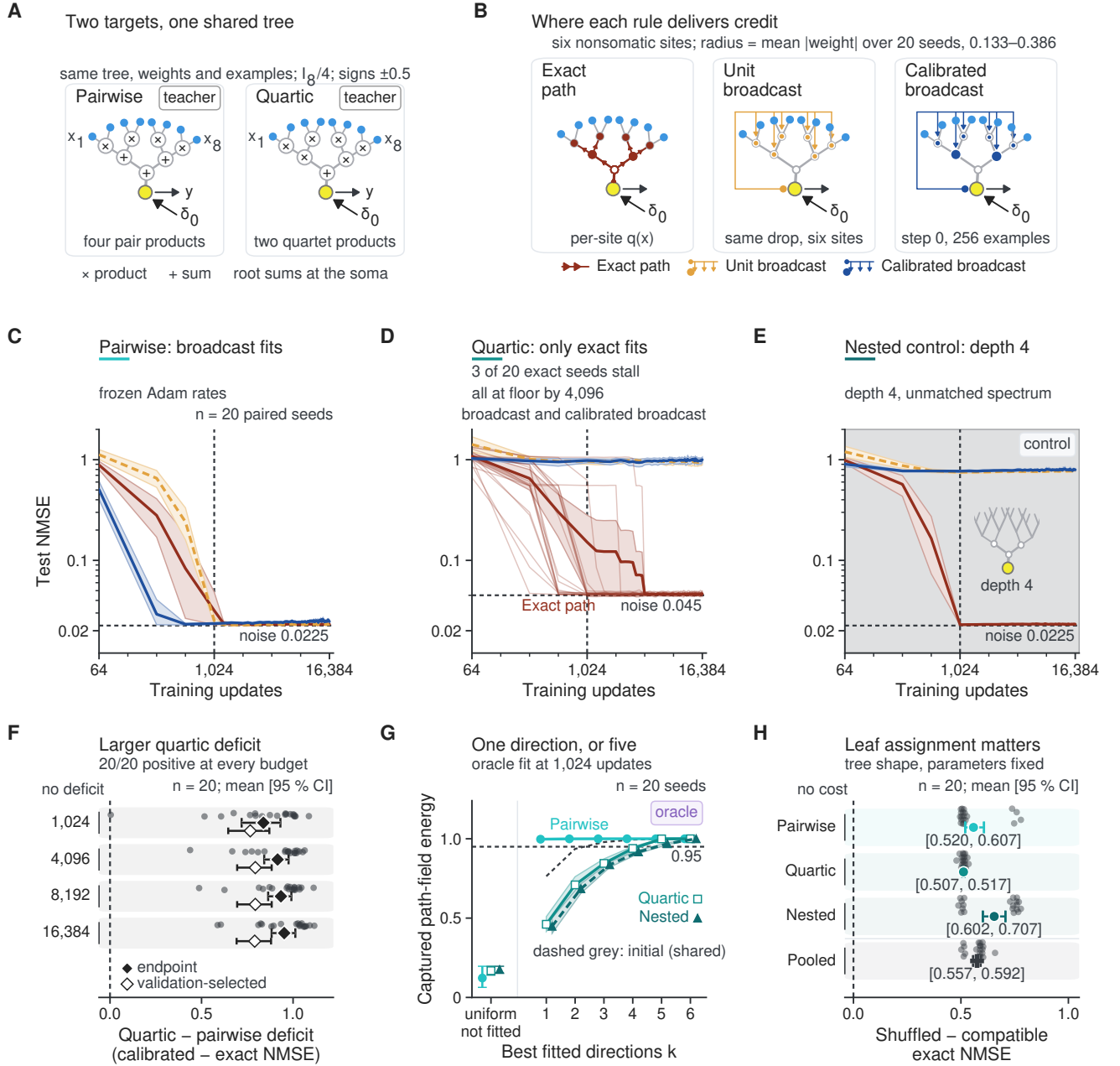


Figure 4: Input-spectrum-matched targets develop different credit geometry and different tolerance of calibrated broadcast credit. **A**, Pairwise and quartic targets on one shared balanced tree: soma lowest, inputs x_1 – x_8 (outer two labelled), both cards badged teacher, emitting the target output y ; the somatic error δ_0 at each soma is the student's, and only the junction badges differ. Schematic, no data. **B**, The three credit rules over the six nonsomatic sites: exact path (per-site $q(x)$), unit broadcast (amber bus) and calibrated broadcast (blue bus; drop-dot radii are the mean absolute calibrated weight, 0.133–0.386, signs mixed). Schematic, except the radii. **C**, Held-out NMSE against updates on the pairwise target at the frozen rule-specific rates; label-noise floor dashed; the vertical dashed line in **C**–**E** is the 1,024-update primary checkpoint used in **G** and **H**. **D**, As **C** for the quartic target, all twenty exact trajectories drawn. **E**, A depth-four nested control on its own tree, not sharing $l_8/4$. In **C**–**E**, $n = 20$ paired seeds with 95% percentile bootstrap bands at fixed checkpoints; the three panels share one axis, but their NMSE denominators are the clean target variances (1.0, 0.5, 1.0), so floor and curve heights are not comparable between panels; the teal ramp orders targets within this figure only. **F**, Quartic minus pairwise difference in calibrated-broadcast minus exact NMSE at four budgets; dots the paired seeds, filled diamonds endpoint and open diamonds validation-selected states ($n = 20$ seeds, 95% paired percentile intervals). **G**, Cumulative exact-trained path-field energy captured by the best k oracle directions, the uniform profile a separate unfitted category, series dodged in k against overprinting. Grey dashed, the step-0 initial spectrum shared by the pairwise and quartic trees; horizontal dashed, the 0.95 threshold; the nested mean is dashed ($n = 20$ seeds, 95% percentile intervals, at that checkpoint). **H**, Exact-path NMSE after shuffling leaf input assignments at fixed shape and parameter count; circles, one task family each; dark plus, pooled over the three families (0.575 [0.557, 0.592]). Grey dots the paired seeds, whose pairwise and nested distributions are bimodal, so those means are not typical seeds; each row prints its interval under its marker ($n = 20$ seeds per target, 95% paired percentile intervals, endpoint state). Source Data: source_data/curated_publication/figure_04_plotted.csv.

Twenty new paired seeds tested the rule under a protocol frozen after exploratory results. All nine rules ran at three common rates through 16,384 updates; the primary rate and 4,096-update window were fixed before these outcomes. On opposed targets, mean held-out NMSE at validation-selected states within the primary window was 2.18×10^{-5} for exact credit, 2.33×10^{-5} for the local distal gate, 2.28×10^{-5} for the two-profile oracle with unit proximal credit, and 0.487 for broadcast; the fixed-checkpoint broadcast trajectory reaches 0.522 at the 4,096-update rule of Fig. 5D. On aligned targets, the corresponding exact, local-gate and broadcast errors were 7.53×10^{-6} , 1.06×10^{-5} and 5.53×10^{-4} . The opposed-minus-aligned difference in broadcast-minus-gate error was 0.487 (95% paired interval, 0.360–0.615), positive in all twenty seeds. It remained 0.468 (0.332–0.604) within the longer window (Fig. 5E).

The controls identify what the gate must preserve. Reversing its distal selection gave opposed-task NMSE 0.949, with four of twenty fits never improving on their initialization. Applying it at proximal compartments as well gave 0.123 and prevented both proximal inhibitory gains from learning in every seed: their eligibility is nonzero only during inhibition, precisely when that gate closes, and the validation-selected state came within 256 updates, its fixed checkpoint reaching 0.553 (Fig. 5F). The proper local rule changed all 24 parameters and succeeded across the three tested rates. The continuous, shunt-proportional shape of the same local distal gate reached NMSE 2.32×10^{-5} under the same primary rate and window, within 1.4×10^{-6} (-1.7 to 4.8×10^{-6}) of the exact path (Fig. 5F). Both gates were tested with binary inhibitory contexts; continuously varying or noisy context remains untested.

The gate coarsens the exact rule’s strong context-dependent gain ratio. At matched weights, broadcast mixes opposing context-conditioned updates despite a nonnegative per-example inner product with the exact gradient in every seed: in an earlier twenty-seed cohort the fraction of the distal gradient surviving context averaging was 0.64 for exact credit at both initial and broadcast-trained weights, but fell from 0.33 to 0.17 for the calibrated broadcast (Fig. 5G). This supplies a local implementation given inhibitory context and somatic error, while leaving their biological origin open. The comparisons remain window- and parameter-bounded. Exact-path and local-distal-gate fits reached no conductance bound, whereas eighteen of twenty opposed broadcast fits had done so by 4,096 updates and all twenty by 16,384; the earlier wide-bound and rate controls remain in Supplementary Fig. S20.

Task organization sets the benefit; budget sets the ranking

The preceding comparisons hold a compatible tree fixed. Varying serial structure instead tests whether successive compartments help remove unwanted multiplicative changes in input strength. Physical depth D_p counts forward stages, whereas feedback resolution counts distinctions in the backward field.

Each network contained 64 somata and eight nonsomatic compartments per soma. D1 [8], D2 [2,3] and D3 [2,1,2] distributed those compartments across one to three stages in the three- and four-tier comparisons while preserving contacts and trainable resources. The two-tier D2 tree used [4,1], also with eight compartments. Branching vectors are ordered from soma to tips. A resource-identical grouped-point control connected all eight modules directly to the soma; raw-additive trees removed divisive conductance normalization. Flexible point networks provided additional, approximately parameter-matched comparisons (the ceiling of Fig. 6D).

Distal excitation carried class information multiplied by nuisance gains. Inhibitory sensors reported those gains with sensor fidelity α : zero retained the sensor distribution but removed trial-specific information, and one reported the factor exactly. With inhibitory reversal zero, the sensors entered the conductance denominator and could support successive divisions [24, 41]. Nested factors acted over fine, coarse and global input groups; flat factors had equal-resolution supports; locally cancellable factors placed the needed ratio within each module (Fig. 6B). Both nested and flat targets used commutative products; they differed in where factor information was available.

Task-family controls showed what serial organization contributed: at full sensor fidelity, exact-BP serial trees exceeded the resource-identical grouped-point control by 30.32 percentage points for nested gains (30.02–30.62) and 22.20 points for flat gains (21.73–22.65), and by exactly zero for locally available ratios, where all ten pairs were numerically identical. Removing trial-specific sensor information eliminated these benefits (Fig. 6C). Within the nested three-tier task, aligned exact-BP accuracy rose from 61.0% at D1 to 67.5% at D2 and 91.8% at D3, whereas the resource-identical grouped-point control stayed at 61.0% at every depth, reversed sensor placement stayed between 60.6 and 61.1%, and raw-additive trees fell to 51.9%; a parameter-matched point network reached 99.0% (Fig. 6D). Adding a fourth stage did not help: in the four-tier task D4 minus D3 was -1.46 points (-1.74 to -1.16), negative in all ten seeds, and the two-tier task was tested only at D1 and D2 (Supplementary Figs. S22B,C and S24B). Serial structure was useful when the task required integrating factor information distributed across modules.

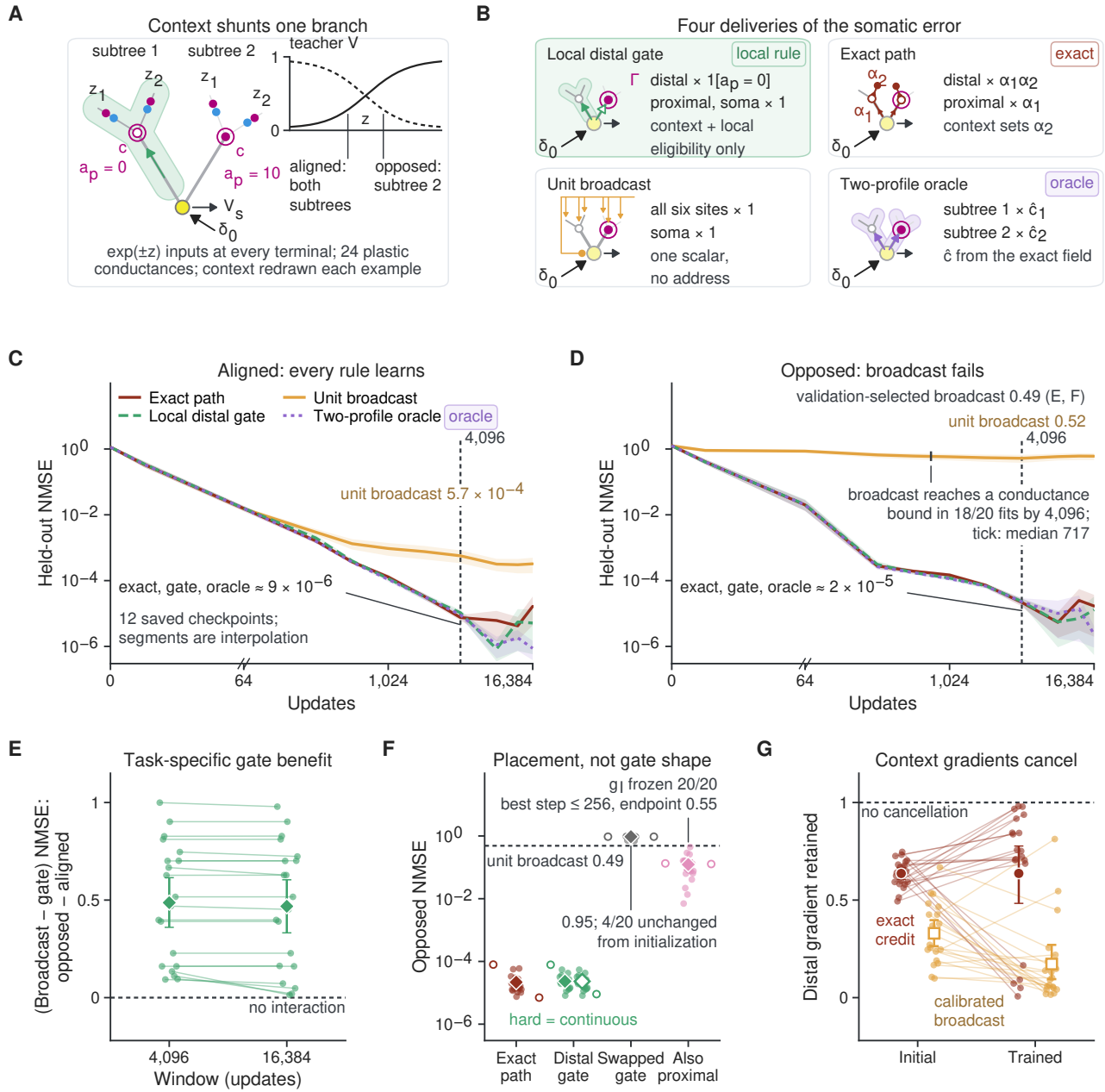


Figure 5: A local inhibitory gate delivers branch-selective credit that matches exact path credit in a conductance tree. **A**, Seven-compartment circuit: four terminals driven by $\exp(\pm z_j)$ feed two junctions and a soma that δ_0 enters; binary context raises one junction's inhibitory conductance ($a_p = 10$, closed, descendants attenuated). Inset, the nominal unperturbed teacher voltage, aligned solid and opposed dashed; fits used independently perturbed teachers. Blue, excitatory contacts; magenta, inhibitory contacts and the context gate. Schematic, except the inset. **B**, Four deliveries of δ_0 on identical trees: local distal gate ($\times 1[a_p = 0]$ at terminals, unit above), exact path (α factors), unit broadcast and two-profile oracle. The card draws the hard gate; both gates restrict Γ (Eq. 4) to $\{0, 1\}$ and $(0, 1]$ and both appear in **F**. Schematic. **C, D**, Held-out NMSE for aligned and opposed targets; means of 20 paired seeds with 95% pointwise bootstrap bands; rule key in **C** (exact path solid, gate dashed, oracle dotted, unit broadcast solid); tick in **D**, the median first bound contact at 717 updates. Exact path, gate and oracle coincide within the line width from 256 updates to the 4,096-update rule, and separate beyond it. **E**, Opposed-minus-aligned difference in broadcast-minus-gate NMSE at both windows, hairlines joining seeds; dashed rule, no interaction; $n = 20$ paired seeds, 95% paired whole-seed bootstrap intervals at validation-selected states, sign-flip $P = 3.8 \times 10^{-6}$ (Holm). **F**, Opposed NMSE in the 4,096-update window by gate shape and placement; at the distal-gate tick the filled and open diamonds are the hard gate $1[a_p = 0]$ and the continuous gate $1/(1 + g_p^I a_p)$, both with filled seed clouds; small open circles, each column's own rule at Adam 0.01 (left) and 0.1 (right); dashed rule, unit broadcast. At the swapped gate spread and intervals are narrower than the markers; at the two shapes only the intervals are. $n = 20$ seeds per condition and rate, 95% whole-seed bootstrap intervals, validation-selected states. **G**, Earlier cohort: distal gradient retained after context averaging at initial and broadcast-trained weights, hairlines joining each seed; dashed rule at 1, no cancellation. Exact-credit means coincide at the two states but the trained distribution splits, five of twenty below 0.17. $n = 20$ seeds, 95% whole-seed bootstrap intervals. Source Data: source_data/curated_publication/figure_05_plotted.csv.

Credit comparisons within this useful forward architecture depended on training duration. Ten paired seeds compared D1 exact BP with five D3 credit/optimization conditions for up to 600 epochs, retaining the original rates, parameter groups and thirty-epoch validation patience. Because optimizer and random states were not archived, these are restarts of the original configurations and seeds rather than checkpoint continuations, which is why the 180-epoch D3-minus-D1 gain reads 30.82 points here and 30.86 in the original-budget fits (Supplementary Figs. S22–S24 and Table S9). At validation-selected states from the same trajectories, exact-BP D3 exceeded D1 by 30.82 points at 180 epochs and 38.17 at 600 (Fig. 6E). Eight of ten D1 references stopped early, whereas all fifty D3 fits reached the cap with validation loss still falling (Fig. 6F). The forward benefit is therefore budget-specific, with unequal optimization maturity across depths.

Within D3, the LocalCA exact-path-minus-shared-soma accuracy gap changed from +10.86 points at 180 epochs to −1.52 at 600 (95% paired interval, −1.83 to −1.23), with this budget-indexed sign change in all ten seeds. The mean gap remained negative from epoch 315, reached −2.94 points at epoch 486, and moved back towards zero by 600; the pointwise interval straddles zero intermittently between epochs 300 and 325, so the crossing epoch is not resolved (Fig. 6G). Exact-path LocalCA nevertheless retained lower mean cross-entropy over epochs 180–600, its advantage decaying from −0.073 to −0.021 nats and ending at 0.254 versus 0.276, negative in nine of ten seeds at 600 (Fig. 6H). Accuracy counts threshold crossings, whereas cross-entropy also reflects predicted probabilities. Under the BP recipe, the 600-epoch exact-minus-broadcast accuracy gap was 0.48 points (0.40–0.55), positive in all ten seeds (Fig. 6E). Thus serial organization and spatial credit answer distinct questions: distributed factor information supports the forward architecture, while the eventual ranking of the tested credit rules remains unresolved (Supplementary Figs. S22–S24 and Table S9).

Reconstructed arbors provide spatial routes beyond a shared broadcast

The learning tasks motivate asking what delivery patterns real arbors supply. MICrONS links reconstructed anatomy and connectivity to visual responses [42]. Observed skeletons, soma positions and mapped contact locations and areas define the geometry; conductances, voltages and gradient fields are modeled.

Each inhibitory-bearing segment supplied a candidate route over its input-bearing descendants (Fig. 7A). These supports form dictionary columns, supplemented by a constant column for common broadcast; in the median-sized initial cell the seven selected routes nest and cover 32 of its 70 input-bearing sites (Fig. 7B). For a site-by-route matrix A_K , diagonal excitatory-area weights W and weighted least-squares projector $P_{A_K, W}$, capture of a field $\mathbf{t}$ was

$$\mathcal{C}(A_K; \mathbf{t}) = \frac{\|P_{A_K, W} \mathbf{t}\|_W^2}{\|\mathbf{t}\|_W^2}, \quad \|\mathbf{v}\|_W^2 = \mathbf{v}^T W \mathbf{v}. \quad (7)$$

For an ensemble of perturbations, numerator and denominator sum over field columns. The field is the modeled gradient change after one small focal shunt, and capture is the fraction of its weighted energy inside the dictionary span (Fig. 7C). A common-constrained SVD dictionary provided a ceiling at the same column budget. Controls used random anatomical routes, cable-distance bins, shuffled routes preserving nonzero counts, and surrogate trees preserving degree and depth while changing ancestry.

Small focal shunts in reciprocal cables generated gradient-change fields without inspecting the selected dictionary. These fields nevertheless favor ancestry by construction under a somatic loss, as derived below. The test asks how a small dictionary compresses the tree’s own passive responses. In the initial eight-cell cohort, broadcast captured 31.4% of field energy, near the 32.6% rank-one ceiling (Source Data). Every dictionary therefore contained one common broadcast and $K - 1$ spatial profiles, with the protocol fixed before evaluation in 47 disjoint cells from the same mouse and a reconstructed second mouse.

Varying K separates the common signal from the additional spatial information each dictionary can capture (Fig. 7D). At eight columns in the disjoint 47-cell cohort a shared broadcast captured 0.203 of total energy and broadcast-plus-ancestry 0.619, against 0.578 for degree–depth matched surrogate trees, 0.511 for depth bins and 0.436 for shuffled routes (Fig. 7D); the spatial share beyond that broadcast is 0.416 for ancestry against 0.376, 0.309 and 0.233 for the three controls (Fig. 7E). Cell by cell, ancestry exceeded its own surrogate mean in 38 of 47 cells, but in 11 cells at least half of the 200 surrogate trees matched or exceeded the actual tree (Fig. 7F). The paired ancestry-minus-surrogate advantage was 0.041 of total energy (95% cell-bootstrap interval, 0.020–0.062) and 5.08 percentage points of post-broadcast energy (2.44–7.68); against depth bins it was 0.108 (0.077–0.135) and 13.51 points (9.86–16.84), positive in 45 cells, with



Figure 6: Task organization sets the forward benefit of serial dendrites, and the training budget sets the credit-rule ranking. **A**, Eight nonsomatic compartments per soma in one stage (D1 [8]) or three (D3 [2, 1, 2]): blue, the excitatory class-bearing contacts; carmine, inhibitory sensors, all tiers in D1 and one per stage in D3; δ_0 , somatic error; footer, the grouped-point control. D2 and D4 keep that inventory. Schematic, no data. **B**, Generative model: nested gains act on 4, 2 and 1 blocks of the stream, flat gains on eight blocks, local-ratio inputs inside every module; sensors report tier ℓ at fidelity α . Schematic, no data. **C**, Serial-minus-grouped-point accuracy at fixed D3 under exact BP, by task family and fidelity (colours ordinal here); the local-ratio value at $\alpha = 1$ is an exact tie, not a sampled null (LocalCA counterpart, Supplementary Fig. S22D). **D**, Three-tier task: accuracy against serial depth for three credit arms, a raw-additive mechanism control, two resource-matched controls (grouped and reversed, dodged in depth, agreeing within 0.5 points) and the point-network ceiling. Intervals are smaller than the markers except for the additive arm, whose means shift by up to 1.5 points across re-runs; the four-tier cohort is separate and not drawn. **E**, Validation-selected accuracy for six 600-epoch restarts of the same seeds (not continuations); dotted, each rule's autograd-broadcast variant, the local one overprinting shared soma; thin grey, the D1 reference, lightened once fewer than eight of its ten fits run; bands exceed the stroke mainly before epoch 100. Paired labels name arms whose 600-epoch ends differ by 0.48 and 0.02 points. **F**, Best validation loss for the same arms; open markers, D1 stopping epochs. **G,H**, Paired exact-path-minus-shared-soma differences in accuracy and cross-entropy; the band in **G** is the drawn 300-330 window over which the crossing is unresolved; the dashed rule in **E-H** is the 180-epoch budget. Points are means over ten paired seeds; bars and shading are descriptive 95% seed-bootstrap intervals, of each arm's mean in **D-F** and of the paired difference elsewhere, pointwise in **E-H**; accuracy endpoints are validation-selected. Source Data: source_data/curated_publication/figure_06_plotted.csv.

shuffled and random routes at 23.06 and 27.17 points (Fig. 7G). Actual ancestry improves average compression within this passive response family, while coarse geometry explains much of the capacity.

Capture and delivery cost are separate constraints (Fig. 7G, right columns and footer). The eight-column ancestry dictionary used 18.75% of the entries of a fully dense dictionary on average—12.5% for the always-dense broadcast column and 6.25% for the seven route columns—and its shuffled-routes control preserved exactly the same nonzero count at much lower capture. Equal column count does not ensure equal rank, and nonzero entries are a delivery-cost proxy rather than feedback synapses or metabolic expenditure.

The ancestry-over-surrogate-over-depth ordering held in the initial eight-cell cohort and in the Pinky second-mouse cohort, where eight of ten qualifying cells supported the eight-column comparison (Fig. 7H); the rank of the random-route control, by contrast, exchanged with depth bins between cohorts. Pinky’s 9–13 input-bearing sites put ancestry near the residual-capture ceiling, so lower-budget comparisons and exclusions remain separate from the larger arbors. The initial and disjoint cohorts come from one MICrONS mouse; Pinky is a second animal. Geometry, labeling and compression sensitivities, including direct-label coverage and direct-versus-proxy agreement, accompany the cohort controls in Supplementary Figs. S25–S28, with the wiring-density reconciliation in Supplementary Fig. S25A, the coarse-energy controls in Supplementary Fig. S27B,C and the preprocessing audit in Supplementary Fig. S28A; there, no inhibitory-targeting endpoint persists after joint three-dimensional matching (Supplementary Fig. S26B–D), and the initial cohort’s anatomy-specific coarse-energy excess does not replicate in the disjoint 47-cell cohort (Supplementary Fig. S27D).

Focal shunting sets ancestry-defined route gains with selectivity set by electrotonic state

A focal shunt provides an exact link between anatomical routes and conductance-dependent gains. We compared a focal inhibitory shunt with a current injection matching its baseline focal current in reconstructed reciprocal cables. A separately solved somatic current restored baseline somatic voltage after either intervention (Fig. 8A). The endpoint was the change in the modeled excitatory-conductance gradient at open excitatory contacts.

Let G be the passive conductance matrix and $\mathbf{b}$ the current-source vector, so that baseline voltages satisfy $G\mathbf{V} = \mathbf{b}$. With $\mathbf{e}_k$ the unit vector at compartment k , adding inhibitory conductance κ_k at compartment k changes both G and its source: $G' = G + \kappa_k \mathbf{e}_k \mathbf{e}_k^\top$, $\mathbf{b}' = \mathbf{b} + \kappa_k E_1 \mathbf{e}_k$. The matched injection supplies baseline current $\kappa_k (E_1 - V_k)$ without changing G . The adjoint $\mathbf{q}$ gives loss sensitivity to current injected at each compartment. For a fixed somatic loss derivative, it changes by

$$\mathbf{q}' = \mathbf{q} - \frac{\kappa_k q_k}{1 + \kappa_k (G^{-1})_{kk}} G^{-1} \mathbf{e}_k. \quad (8)$$

The column $G^{-1} \mathbf{e}_k$ gives the voltage response to unit current at k . On a passive tree with a somatic loss, the resulting adjoint change has an exact ancestry structure: all descendants of the shunt share the attenuation factor $[1 + \kappa_k (G^{-1})_{kk}]^{-1}$, while each sister block along the path to the soma shares its own, and the soma-side remainder a fifth factor (Fig. 8B; Supplementary Section S8). These disjoint blocks form an ancestry partition. Weighting their indicators by the baseline soma-to-site transfer gives a dictionary on which the shunt acts through diagonal gains, providing a physical instance of Γ_μ , drawn as the block indicator matrix and its gain column in Fig. 8B. The diagonal form applies in this partition basis; it need not hold for an arbitrary selected set of overlapping routes.

In the small-perturbation fields used for Fig. 7, the driving-force term was substantial (0.99–3.50 times the full field’s weighted squared energy across the initial eight-cell cohort) and opposed the adjoint term (weighted correlations -0.979 to -0.913), yet 97.1–99.9% of full-field energy remained within each focal ancestry partition (Supplementary Section S8).

We compared descendants $D(k)$ with depth-matched off-route sites $C(k)$; Supplementary Section S8 gives a kernel criterion for stronger absolute adjoint changes below a shunt, with the within-cell control in Supplementary Fig. S29B and the contact-label and reversal-potential controls in Supplementary Fig. S29E,F. The analyzed gradient was sensitivity to open excitatory conductance, $\gamma_i = q_i (E_E - V_i)$. The local driving force $E_E - V_i$ can change differently within each block, so the complete synaptic gradient need not share a single block gain. With primes denoting post-intervention values and ε_γ a numerical floor defined in Methods, we measured absolute log change, $m_i = |\log[(|\gamma'_i| + \varepsilon_\gamma)/(|\gamma_i| + \varepsilon_\gamma)]|$, and localization $\Lambda_k = \text{median}_{D(k)} m_i - \text{median}_{C(k)} m_i$. Positive localization means a larger gradient change below the shunt.

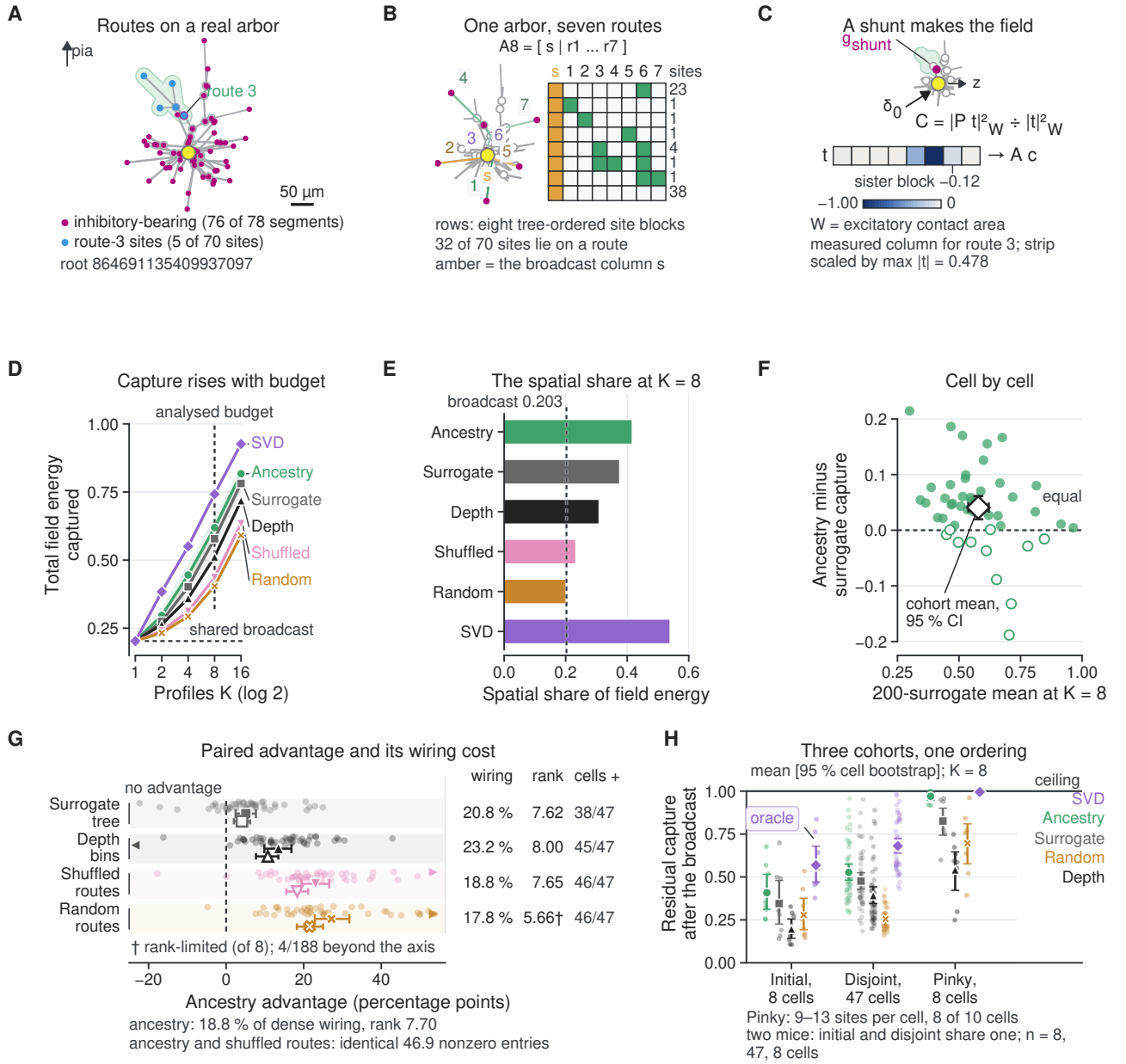


Figure 7: Ancestry routes on reconstructed arbors compress a cell's own focal-shunt response fields better than four budget-matched controls, at a fifth of dense wiring, and the advantage is modest and heterogeneous. **A**, Median-sized arbor of the initial eight-cell cohort (root 864691135409937097, 78 segments): its 76 inhibitory-bearing segments and route 3 over the five sites it addresses; schematic, no data; scale bar, 50 μm . **B**, The same arbor's seven $K = 8$ routes and the matrix A_8 , collapsed to eight tree-ordered site blocks; supports nest; schematic, no data (entries measured). **C**, A focal shunt on route 3 attenuates its descendants and makes the field t (block means); capture is the W -weighted energy of $P_{A_K, W} t$ as a fraction of that of t ; schematic, no data (field column measured). **D**, Total captured energy against budget K ; cell means with unpaired 95% cell-bootstrap bands for ancestry and surrogates; dashed floor, the shared broadcast 0.203; $n = 47$ cells, 46 at $K = 16$. **E**, The spatial share of that energy at $K = 8$, i.e. capture beyond the shared broadcast, for the six dictionaries; dashed line, the shared broadcast 0.203, shown for scale and removed from every bar; cell means, no interval; $n = 47$ cells. **F**, Each cell's ancestry capture minus its own 200-surrogate mean, against that surrogate mean; dashed line, zero difference; open green circles, the 11 cells whose surrogates match the actual tree at least half the time; open black diamond, the cohort mean, with 95% cell-bootstrap intervals on both coordinates; $n = 47$ cells at $K = 8$. **G**, Paired ancestry-minus-control advantages at $K = 8$ on post-broadcast (filled) and total (open) scales, with all 47 within-cell differences; means and 95% cell-bootstrap intervals, four of the 188 differences clipped to the axis edge; right: wiring, rank, positive cells. **H**, Residual capture after the broadcast in three cohorts from two mice (the initial and disjoint cohorts are one MICrONS mouse, Pinky a second); means with 95% cell-bootstrap intervals at $K = 8$; dots, individual cells; $n = 8, 47$ and 8 cells. Typed-contact and joint 3D matching controls are in Supplementary Figs. S25B and S26B–D. Fields are modeled passive responses: compression capacity, not observed teaching. Source Data: [source_data/curated_publication/figure_07_plotted.csv](#).

In the permissive normalized model, shunts changed descendant gradients more strongly than depth-matched off-route gradients, with increasing localization at higher dose (Fig. 8C,D). Current injection changed descendant gradients roughly threefold less. Its unrelated-site changes were zero by construction under the soma-restoring control, and its depth-matched off-route changes averaged 2.7×10^{-4} at unit dose (cells to 8.8×10^{-4}); this is not a general claim that current injection cannot change a gradient. Holding the post-shunt voltage but replacing its adjoint with the baseline adjoint removed most localization, supporting a transport contribution beyond the voltage change alone (Fig. 8E). Adding the post-shunt adjoint on top of the substituted driving force ($q'd' - d'$) added 0.079 log units of localization, positive in all eight cells, while substituting the driving force alone left 0.026 against 0.104 for the full shunt.

Physical calibration constrained the spatial selectivity of the gradient change. At axial resistivity $R_a = 150 \Omega\text{cm}$ and specific membrane resistance $R_m = 15,000 \Omega\text{cm}^2$, mean shunt-minus-current localization was small and opposite-signed in the initial eight cells and a disjoint 45-cell cohort from the same mouse: -0.0019 and $+0.0024$, respectively. Signed changes showed attenuation of both descendant and off-route gradients at high resistance; lowering resistance primarily increased descendant attenuation (Fig. 8F), whereas the matched injection enhanced them ($+0.0099$ at $R_m = 15,000$, $+0.043$ at $R_m = 300$, initial eight cells). This increased the localization contrast by more than an order of magnitude (Fig. 8G). Distributed background conductance produced the same qualitative recovery at fixed high membrane resistance (Fig. 8H; -0.0004 , $+0.121$ and $+0.259$ at background leak zero, one and four times baseline, positive in eight of eight cells at one and four), consistent with a dependence on electrotonic state [25, 43].

Adding weak voltage-dependent channel currents produced a dose response close to the passive response at $R_m = 1,000 \Omega\text{cm}^2$ with one baseline-leak unit of distributed background conductance (Supplementary Section S8; Fig. S30D,E). This checks the local linearization around solved steady states. The baseline Jacobian included gate derivatives but remained fixed across doses, so the check does not establish robustness to strongly active dendrites or channel kinetics.

The partition identity and electrical calibration connect anatomical dictionaries to conductance-dependent route gains [26]. Their selectivity depends on cable state. These imposed perturbations establish modeled transport, not endogenous error production or plasticity.

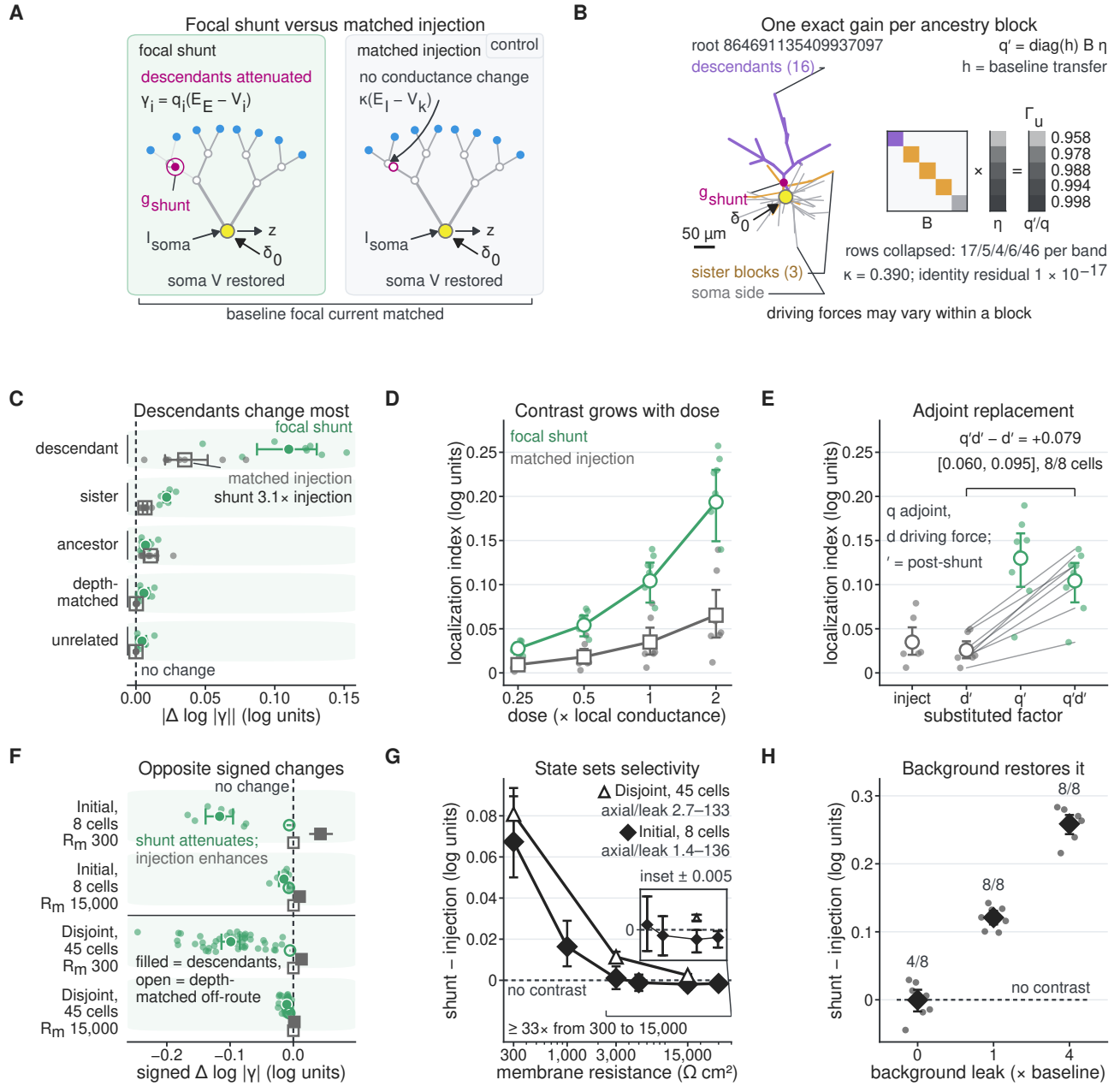
Measured responses bound the anatomical alignment proposal

Measured responses test whether functional relationships follow the ancestry structure suggested by modeled capacity and transport. Seven MICrONS targets had complete arbors and imaged partners: 69 partners in the selected-scan dataset and thirteen eligible scans in the scan-complete analysis. These recordings sampled stimulus responses rather than learning, so no credit was delivered (Fig. 9A).

Partners with more shared dendritic ancestry did not have reliably more similar condition-mean responses after controlling Euclidean separation and soma-to-contact path-length differences (shared-path comparison, Fig. 9A; estimate, Fig. 9B). Mean partial rank correlations were -0.069 for selected scans and -0.048 for all eligible scans, the same seven targets re-analyzed, and both target-bootstrap intervals included zero (Fig. 9B,C). The null did not depend on the measure: raw shared-path (-0.025), tree-distance (-0.017) and same-major-branch (-0.003) associations were also centered on zero (Fig. 9C; Supplementary Fig. S31). This provides no evidence of preferential ancestry alignment in the sampled partners, at a different scale from population-level like-to-like wiring [44].

To establish the sensitivity of this small sample, response-level simulations preserved observed partners, shared inputs, repeated stimuli and seven-target inference, with noise calibrated to measured repeat reliabilities. Under the specified Gaussian ancestry-signal model, 80% detection required a mean simulated target-level partial rank correlation of approximately 0.25 (0.249 by interpolation, or $\lambda = 0.65$ once the Monte Carlo interval is respected), and the observed estimate and its whole interval lie below that point (Fig. 9D). With perfect reliability the same threshold falls only to 0.248, so the limit is the seven-target sample rather than measurement noise. The observed repeat reliabilities used for attenuation vary across recordings (Fig. 9D, inset). This conditional threshold leaves weaker effects, different signal geometries and unobserved partners unresolved (Methods; Supplementary Section S9).

A separate passive-tree model predicted each target's responses from its recorded partners. Its selected ancestry routes reached only 48.0% of mapped inputs, and in six of the thirteen scans every route reached a single input (Fig. 9E,F), so restricting the transfer profile to these supports tests transfer geometry more than example-dependent credit, and no measured positive control exists for this comparison. Held-out prediction, input-removal and reconstruction controls are grouped in SI (Supplementary Figs. S31–S32), where the ancestry-minus-ridge difference (0.029; 95% target-bootstrap



interval, -0.013 to 0.074) is matched by random anatomical routes (0.031 ; -0.009 to 0.068). This model diagnostic and the measured ancestry-similarity test constrain different parts of the anatomical proposal.

Discussion

Learning benefits from spatial selection when a common signal mixes incompatible synaptic changes. Dendritic morphology supplies candidate delivery patterns, teaching mechanisms supply their coefficients, and conductance regulates their gains. In the single-layer image classifiers, preserving neuronal identity supplies most of the benefit, including with random between-neuron feedback. Contextual conflict favors branch selection, supplied through feedback or local eligibility gating. Hierarchical distractors favor ancestry at a bandwidth predicted by the generator's coefficients. These results distinguish useful information from its physical carrier.

The interaction comparison explains why input rank alone is insufficient: pairwise and quartic targets share that spectrum and a compatible tree, yet learn different credit fields with different tolerance of fixed profiles. Spatial direction matters alongside dimension: even a nearly one-dimensional field can be poorly captured by uniform broadcast. These learned-state properties do not rescue the failed initialization selector. Intermediate dictionaries, adaptive profiles and alternative dynamics remain plausible solutions.

Conductance trees connect this distinction to computation: simple credit learns compatible monotonic teachers, whereas opposed tuning under contextual inhibition benefits from selection. A local inhibitory gate supplies that selection without oracle path coefficients, provided proximal credit remains ungated. The task supplies context and somatic error. Forward compatibility is a separate constraint: the scalar multi-affine cut criterion connects input grouping to hierarchical tensor descriptions and supports structure estimation from noisy examples [45–47]. Serial composition likewise helps remove distributed nuisance factors, but the LocalCA accuracy advantage of exact credit changes sign between the 180- and 600-epoch budgets while exact credit retains lower cross-entropy. The gap is still moving at the cap; forward depth and backward resolution need separate interpretation.

Obtaining coefficients remains a limitation beyond this explicit context gate. The separate encoder experiment exposes two obstacles: identifying a route from noisy cues and, when calibration is small, preventing coefficient leakage into incompatible routes even after identification is correct (Fig. 3F). With 256 noise-free cues the soft readout came within 0.31 points of the oracle, so leakage is a calibration cost rather than an intrinsic property of a soft readout. Exploratory hard selection removes it at every calibration size, but leaves a substantial noisy-cue deficit. Exact paths and projected coefficients elsewhere remain information-access controls. An adequate span alone does not ensure useful delivery.

Reconstructed arbors compress their modeled passive response fields beyond a common broadcast. Degree–depth surrogates explain much of this capacity, and the target fields themselves favor ancestry through passive transport under a somatic loss. Testing alignment beyond that construction requires measured fields or a model with different error sources or active dynamics; changing the passive perturbation family alone would not suffice. Focal shunting supplies a biophysical link by reweighting the baseline-weighted ancestry partition. Driving-force changes partly cancel adjoint changes, while cable state sets the spatial selectivity of the resulting gradient. This concerns sensitivity transport, distinct from inhibition that computes errors or regulates forward gain and load [48–51].

The small recorded cohort provided no evidence of preferential ancestry alignment; the offline learning diagnostic chiefly tests an almost fixed transfer profile. A direct experiment would compare plasticity below a focal inhibitory site with electrotonically matched off-route sites while matching somatic voltage, first-order current and outcome signal. Changing conductance state would test the predicted separation between shared and ancestry-selective effects [22, 52, 53].

The framework complements somatic prediction, segregated teaching, interneuron-mediated errors and burst plasticity [32, 34–36], alongside cable-learning and engineered gating models [39, 54–57]. Spikes, temporal eligibility, recurrence and channel kinetics may change the available profiles and coefficient estimation [58, 59]. The organizing question remains which task-relevant distinctions the arbor and learning mechanism can jointly preserve.

Methods

The Supplementary Information's contents and cohort index (Table S11) locate complete derivations and protocols by scientific question. Resolved configurations, seeds, checkpoints, exclusions and source hashes accompany the numerical

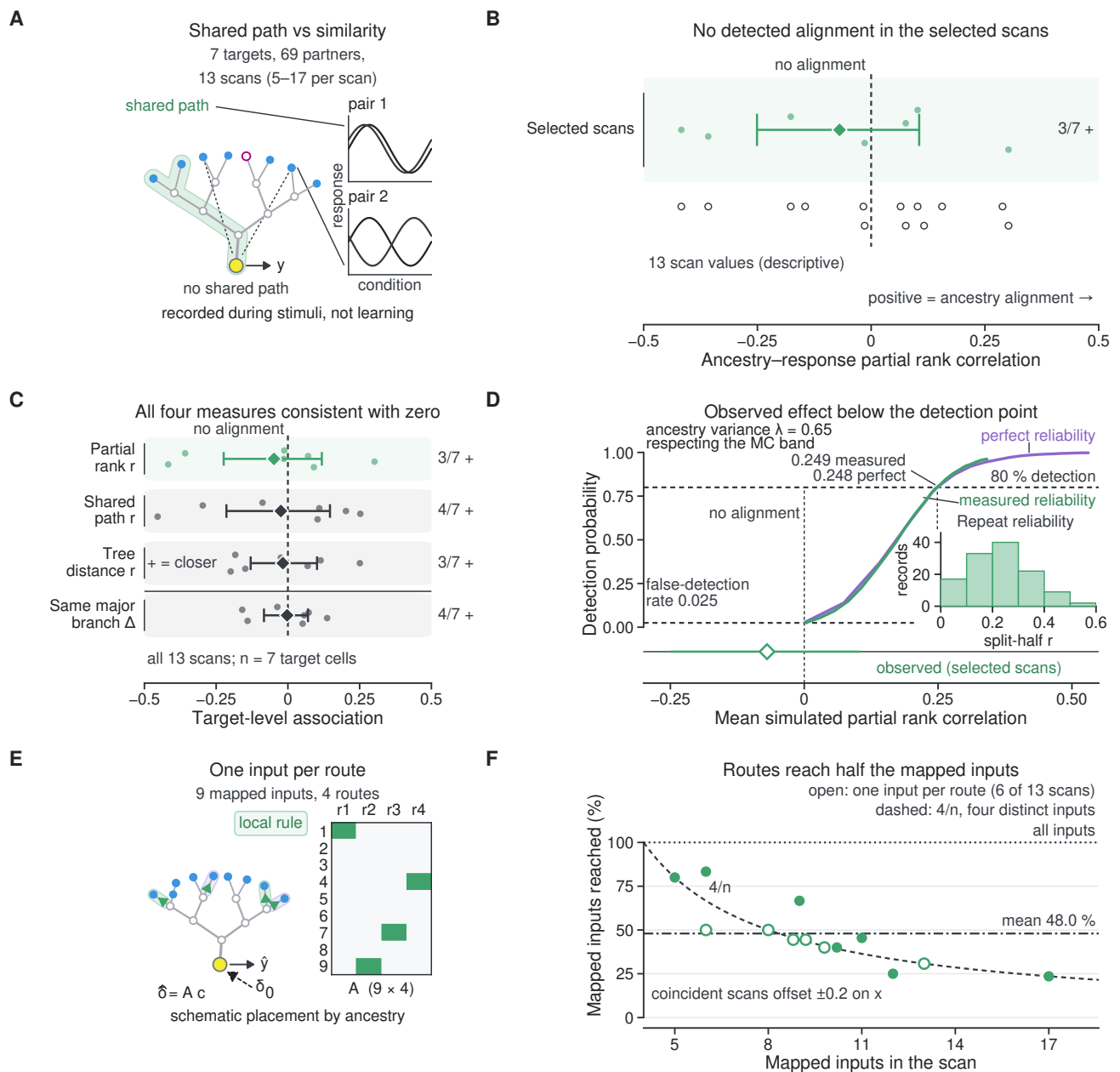


Figure 9: Recorded partner responses show no dendritic-ancestry alignment, and this sample detects only effects above a partial rank correlation of about 0.25. **A**, The comparison: two contact pairs on one arbor, one sharing a soma-to-ancestor path (green patch), the other linked to the soma by dashed hairlines; filled blue discs, excitatory contacts; open magenta ring, an inhibitory partner; insets, their responses. No data: stimulus recordings, no δ_0 . **B**, Ancestry–response partial rank correlation, selected scans; dots, the seven target values; open circles, the thirteen all-eligible-scan values (same targets, stacked where they would overlap); diamond, mean with 95% target-bootstrap interval ($n = 7$): $-0.069[-0.251, 0.106]$. **C**, Same inference, four topology measures, all thirteen scans ($n = 7$; 95% target-bootstrap intervals). Top row, the all-eligible-scan partial rank correlation, $-0.048[-0.224, 0.118]$. Tree distance is sign-flipped, so positive means closer. Below the rule, same-major-branch Δ is a difference in mean similarity, not a correlation; its narrower interval reflects smaller units. **D**, Detection probability against the frozen model’s mean simulated partial rank correlation; shaded 95% Wilson Monte Carlo intervals (half-width ≤ 0.016 , narrower than the stroke); the measured-reliability intercept is its false-detection rate, 0.025 (perfect reliability, 0.030). The rug is the observed estimate ($n = 7$, 95% target bootstrap), not power. Inset, split-half repeat reliability, 123 of the 125 partner–scan records (102 partners), the two with negative r omitted. **E**, Representative scan (median mapped-input count, not outcome-selected): nine mapped inputs placed by ancestry beside the realized 9×4 support, $\hat{\delta} = A c$ imposed, not observed (schematic; dashed δ_0). Route-capsule tints aid separability and the *local rule* badge covers the card; neither encodes a per-route property. **F**, Mapped inputs reached per scan; open, the six scans whose four routes each reached one input; dashed, 4/n, coverage when four routes reach distinct inputs. Not a ceiling: two scans fall below (repeated inputs), three above (multi-input routes). Coincident scans are offset ± 0.2 ; $n = 13$ scans from seven targets, descriptive. Source Data: `source_data/curated_publication/figure_09_plotted.csv`, `source_data/functional_topology_all_scans/`, `source_data/credit_first_figures/` (figure_08_sources.json, figure_08_support.csv and figure_08_actual_support.npz).

554 tables. Diagnostics reusing a cohort retain its original independent units.

555 Conductance models and delivery coordinates

556 In a directed tree, $[b_1, \dots, b_D]$ lists branching factors from soma to tips, giving $\sum_{d=1}^D \prod_{r=1}^d b_r$ nonsomatic compartments. A
 557 child c supplies output $a_c = f_c(V_c)$. With leak g_n^L , aggregate open excitatory/inhibitory conductances $G_n^X = \sum_{i \in \mathcal{S}_n^X} g_i^X x_i^X$
 558 for contact class X , and child couplings $g_{c \rightarrow n}^{\text{den}}$, the steady voltage is

$$V_n = R_n^{\text{tot}} \left(g_n^L E_L + G_n^E E_E + G_n^I E_I + \sum_c g_{c \rightarrow n}^{\text{den}} a_c \right), \quad (R_n^{\text{tot}})^{-1} = g_n^L + G_n^E + G_n^I + \sum_c g_{c \rightarrow n}^{\text{den}}. \quad (9)$$

559 Here $x_i \geq 0$ is contact activity, $g_i \geq 0$ its conductance, and E_X its reversal potential. A terminal-to-soma pass evaluates
 560 the directed steady state. Artificial conductance trees use dimensionless leak one and reversals $E_L = E_I = 0$, $E_E = 1$.
 561 Monotone parameter transforms enforce nonnegative conductances. Branch-output nonlinearities, termed reactivations in
 562 the implementation, are identity functions when disabled. Both excitatory and inhibitory contacts learn in the regular-tree
 563 experiments.

564 For a reciprocal cable residual $\mathbf{F}(\mathbf{V}, \mathbf{g}, \mathbf{x}) = 0$, the nonsingular voltage Jacobian J_V gives the adjoint

$$J_V^T \mathbf{q} = \nabla_{\mathbf{V}} \mathcal{L}, \quad \partial \mathcal{L} / \partial g_i = -\mathbf{q}^T \partial \mathbf{F} / \partial g_i. \quad (10)$$

565 With passive current balance, the synaptic gradient is $x_i(E_i^{\text{rev}} - V_n)q_n$. This established sensitivity calculation supplies
 566 the reference for restricted credit [9, 10, 33, 39]. For output $a_n = f_n(V_n)$, activation error $\delta_n^a = \partial \mathcal{L} / \partial a_n$ gives voltage
 567 error $\delta_n^V = f'_n(V_n) \delta_n^a$. Image LocalCA replaces δ_n^a while retaining f'_n and exact local gate derivatives in eligibility. Fixed
 568 activation-space profiles can therefore vary in voltage space; exact supplied fields recover exact selected gradients.

569 The raw-additive control retains the graph but sums signed inputs without the conductance denominator or reversal-
 570 potential driving force. Historical normalized-additive controls additionally standardize voltage across stage coordinates
 571 within each example; Supplementary Section S2 defines those variants. The optional four- and five-factor rules add
 572 stage-level preconditioners and are described in Supplementary Section S1.

573 In $\text{A}\Gamma$, nonzero diagonal gains preserve the dictionary span and oracle capture while changing coefficients and
 574 conditioning (Supplementary Fig. S5B–D). This describes a passive shunt in its baseline-weighted ancestry partition,
 575 not arbitrary overlapping dictionaries. Stage count, feedback rank, external error dimension and state-dependent path
 576 resolution remain distinct (Supplementary Tables S1–S2).

577 Image feedback ladders and checkpoint diagnostics

578 The trained decoder supplies somatic activation errors $\delta_{i,u} = \partial \mathcal{L} / \partial a_{i,0,u}$ for example i and neuron u . Strict scalar feedback
 579 uses their mean, or their largest-magnitude signed coordinate when the mean’s magnitude is at most 10^{-6} , throughout
 580 the core including somatic gates. Per-neuron feedback preserves each coordinate; the exact path additionally applies
 581 local path derivatives. The historical matched-width fallback retained neuronal identity at the 128-wide somatic stage,
 582 affecting 640 of 2,413,312 core parameters. A separately trained strict-scalar control tests that exception (Supplementary
 583 Fig. S2, Supplementary Section S2 and Source Data indexed in Table S12).

584 The new MNIST ladder uses 128-soma [3, 3] shunting and raw-additive networks, with 40 excitatory and 20 inhibitory
 585 contacts per nonsomatic compartment. Within each architecture, rules share initial contact construction, initialization
 586 policy, decoder and data streams. Pixels are divided by 255 and flattened without further normalization or augmentation.
 587 Each seed divides the 60,000 training images into 48,000 training and 12,000 validation examples; all 10,000 official test
 588 images remain held out. Training uses 180 epochs, batches of 256 and the validation-selected state (minimum validation
 589 cross-entropy), without early stopping. Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$ and no weight decay. Original rates
 590 are 0.0015 for input synapses and decoder, 0.0007 for branch couplings and 0.0005 for reactivation gates.

591 The projected $K = 1, 3$ rules average exact activation errors over the supports in Fig. 1C, preserving the field mean
 592 but not necessarily its norm. Both retain exact somatic errors and local derivatives. Decoder-only learning freezes the
 593 initialized core, verified by its hash. Three development seeds compare rate multipliers 0.3, 1 and 3 for every rule and
 594 architecture, giving 108 fits. Validation alone selects rates. Ten fresh paired seeds per architecture and rule give 120

selected-rate and 120 original-common-rate outcomes from 190 unique fits; 50 fits serve both policies. The projected- $K = 1$ addendum was frozen before fresh outcomes. Complete protocols, clipping and excluded implementation checks are in Supplementary Section S2.

Supplementary Section S2 gives the original fifteen-seed MNIST [60] ladder, ten-seed Fashion-MNIST [61] and twenty-seed raw-additive CIFAR-10 [62] protocols; CIFAR uses a 20-soma $[3, 3, 3, 3]$ tree. Original gradient diagnostics appear in Fig. S8A–C. The fifteen-seed direct-feedback-alignment factorial replaces the decoder derivative by a fixed Gaussian matrix scaled by inverse square root of output dimension, paired with exact-readout references. Recipient permutations preserve coordinate values and bandwidth.

Common-checkpoint diagnostics compare rules at identical weights and states. Exact path transport is checked against automatic differentiation. New capture measurements use 2,048 held-out images at initial and exact-trained states; activation- and voltage-space metrics are kept separate. Gradient cosine, error magnitude by depth and mean-subtracted path energy are distinct summaries; none alone certifies learning performance, and their joint behaviour at 120 valid trained checkpoints is in Supplementary Fig. S3E. Historical depth and noise cohorts, including their task-identity and conductance-validity records, are specified in Supplementary Section S2.

Contextual conflict and ancestry routing

The branch-conflict task uses Eqs. 5–6, with independent opposite-class replacements of nonselected Fashion-MNIST inputs at probability χ . Twenty paired seeds cross $B = 2, 4, 8$ with eight doses. Gated-point and BP calculations receive equivalent updates; A deranged route changes recipients at fixed rank and support. A separate two-stream interference cohort (ten paired seeds 4100–4109, $K = 2$ subtree routes, two contexts) trains on context 0, switches to context 1 for twelve epochs and re-tests context 0 on 1,024 held-out examples; the retained loss is reported as context-switch forgetting (source_data/trained_subtree_address/).

Under centered, symmetric class evidence, shared-mode compatibility is $s(\chi) = 1 - 2\chi(B - 1)/B$, with zero at $\chi_c = B/[2(B - 1)]$. Trained crossings are evaluated separately by mean-curve interpolation and each seed’s first sampled at-or-below-chance dose, retaining non-crossings. Checkpoint re-execution reproduces endpoints and compares exact/shared fields at both common and own trained states (Supplementary Fig. S10B–D). The measured counterpart of $s(\chi)$, reported as the initial signed utility, is the Source Data column mean_initial_signed_utility: the initialization-state inner product of the neuron-shared update with the exact update, signed and normalized so that the analytic $s(\chi)$ is its prediction. It is distinct from the nonnegative capture U reported elsewhere.

The ancestry factorial contains eight eight-dimensional blocks. Context c_i and binary label sign s_i are balanced, and inputs are $\mathbf{x}_{i,b} = a(b, c_i)s_i\mathbf{t}_b + \boldsymbol{\epsilon}_{i,b}$, with seed-specific unit teachers $\mathbf{t}_b$ and Gaussian noise $\boldsymbol{\epsilon}_{i,b} \sim \mathcal{N}(0, 1.1^2 I_8)$. The selected coefficient is one; sibling, same-half nonsibling and opposite-half coefficients are $-0.15, -0.45, -0.75$. A leaf permutation changes the task-to-route assignment without changing the streams. For budget K , ancestry feedback has context-by-block entries $\Phi_{cb}^{(K)} = \sqrt{K/8} \mathbf{1}_{[g_K(\pi(c)) = g_K(b)]}$, where g_K indexes contiguous groups and π maps streams to leaves. Thus every row has unit Euclidean norm.

The update is $\Delta \mathbf{w}_b = -(8\eta/N) \sum_i \delta_i \Phi_{cb}^{(K)} \mathbf{x}_{i,\pi^{-1}(b)}$, with N training examples and rate η . Each seed supplies 2,048 training, 2,048 test and 1,024 switch examples. Full-batch descent uses rate 0.3, eighty training epochs and twenty switch epochs. Controls include deranged ancestry, depth-interleaved and random-sparse supports, dense rank- K fields and a training-informed PCA reference. The best matched control is the per-seed post-training maximum over the four non-PCA controls. Integer test counts preserve ties in the named Wilcoxon comparison; Holm correction covers four budgets. Alternative mean-sign-flip and historical BH analyses remain explicitly labeled in the Source Data indexed in Supplementary Table S12. The separate noisy-context encoder and cue-delay experiments are fully specified in Supplementary Section S3.

Update utility and prospective selection

Let the exact gradient be $\boldsymbol{\mu}$ and the delivered update estimator have mean $\mathbf{m}$ and covariance Σ_{route} at the current weights. For an L -smooth loss and a nonnegative step η , expected loss after the update is bounded above by $\mathcal{L} - \eta \boldsymbol{\mu}^T \mathbf{m} + (L\eta^2/2)(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})$. Optimizing this expression gives utility $U = [\boldsymbol{\mu}^T \mathbf{m}]_+^2 / [2L(\|\mathbf{m}\|^2 + \text{tr} \Sigma_{\text{route}})]$, with zero for a zero denominator. Utility certifies one-step progress under the stated moments; it is not a long-horizon ordering.

Synthetic quadratic screens use sixteen Haar coordinates and a designed signal/noise crossover, without a forward neuron (Supplementary Fig. S3). Retrospective factorial utility uses actual full-batch updates and a global logistic-curvature bound; maximum bandwidth matches its selection success. Supplementary Figs. S34–S35 retain the failed 12,800-fit prospective selector, its loss to fixed maximum-budget and rank-only baselines in every seed, and the 25,600-fit dynamics extension with evolving noise, observed-context-count controls, sealed choices; the computational-cost accounting is indexed in Supplementary Table S12.

Interaction compatibility, fixed profiles and learned spectra

For a subtree leaf set S , C_S arranges the target’s Fourier interaction coefficients by products inside and outside S ; C_S° removes the constant inside row. Scalar multi-affine trees represent the target exactly if and only if every centered cut has rank at most one. If $\mathcal{S}(T)$ contains all proper cuts of tree T and singular values are ordered from largest to smallest,

$$\text{MSE}(f, \hat{f}_T) \geq B(T; f) := \max_{S \in \mathcal{S}(T)} \sum_{j \geq 2} \sigma_j(C_S^\circ)^2. \quad (11)$$

Overlapping cut tails cannot be added as a lower bound. Supplementary Section S4 gives the constructive proof, minimum-depth algorithm and its relation to hierarchical tensor representations. The twelve-tree census with its candidate-selection outcome, the noisy-query estimator, end-to-end learning pipeline, conductance grouping controls and Boolean truth tables remain in Supplementary Figs. S12 (selection in panel C), S14–S15, S18 and S36. The theorem’s assumptions do not extend automatically to general conductance ancestors.

Three development seeds select rates with equal budgets; twenty fresh seed blocks pair matching and quartic conditions across trees, assignments, initialization, inputs, additive noise and minibatch order. Exact, unit, calibrated and sign-only profiles share local eligibilities; calibration averages path derivatives over 256 unlabeled initialization examples. Selected/common rates under SGD and Adam yield 432 development and 1,200 fresh fits. Coefficients stay within $[-2, 2]$, with bound contacts recorded. A balanced extension replays all seven original checkpoints through 1,024 steps, then continues all 720 unique selected/common-rate algebraic trajectories to 16,384. Terminal and validation-selected extensions describe the same seeds (Supplementary Section S4). Noisy held-out NMSE divides by clean target variance: one for matching and nested, one-half for quartic. All have noise variance 0.0225. The primary Adam contrast subtracts the pairwise calibrated-minus-exact gap from its quartic counterpart.

For evaluation-example-by-site path matrix Q , the best rank-one profile is the leading right singular vector. Its capture is $\sigma_1(Q)^2 / \sum_j \sigma_j(Q)^2$, an oracle reconstruction with example-dependent amplitudes. Additional diagnostics evaluate uniform, calibrated and ancestry profiles, loss-residual weighting, eligibility-weighted updates and root-mean-square normalization of each site. Participation-ratio effective rank is $(\text{tr}C)^2 / \text{tr}(C^2)$ for uncentered $C = Q^T Q / N$. At fixed weights and inputs, changing labels leaves the path field unchanged; eighty relabeling controls verify this. Independent checkpoint analysis and faithful replay of 600 earlier fits provide numerical and trajectory checks. Complete definitions, pairing audits, optimizer effects and the positive-conductance boundary are in Supplementary Sections S4–S5.

Conductance trees with context-dependent feature tuning

The seven-compartment model uses identity branch outputs in Eq. 9. Two independent $z_j \sim \mathcal{U}[-2, 2]$ generate positive inputs $\exp(\pm z_j)$ for both subtrees. Every terminal has two excitatory and two inhibitory gains; binary context inhibits one proximal compartment. All 24 conductances are positive and parameterized by logarithms. Students start from terminal gains one, proximal inhibitory gains 15, daughter couplings 20 and 0.2 per subtree, and soma couplings four. These proximal and coupling values match the teacher’s nominal scaffold, not its realized weights. Independent Gaussian log perturbations have SD 0.4 in students and 0.15 in teachers. Full generators and eligibility derivatives are given in Supplementary Section S5.

The local-gate follow-up used twenty new paired seeds and nine rules at Adam rates 0.01, 0.03 and 0.1, giving 1,080 trajectories. Each task supplied 2,048 training, 1,024 validation, 4,096 test and 512 unlabeled calibration examples. The primary rate 0.03 was fixed from earlier work, not selected on these seeds. Minibatches contained 128 examples; logarithmic conductances stayed within $[-7, 7]$. Minimum validation NMSE selected states within fixed 4,096- and 16,384-update windows. The somatic error was $(V_s - y) / \text{Var}_{\text{train}}(y)$; NMSE uses clean target variance. At a terminal with parent p , the hard gate multiplied this error by $\mathbf{1}[a_p = 0]$, where proximal inhibitory activity a_p was zero or ten.

Proximal and somatic multipliers remained one. The continuous gate used $1/(1 + g_p^1 a_p)$. Both combined these signals with the same exact local eligibilities and accessed no exact path derivatives.

Controls used exact credit, unit and initially calibrated broadcast, two weighted distal oracle profiles with unit proximal credit, three oracle profiles including a proximal column, reversed distal gating, and gating at proximal sites as well. Two primary paired sign-flip tests—opposed broadcast-minus-gate error and its opposed-minus-aligned difference—were Holm-adjusted together. Other comparisons and 95% whole-seed bootstrap intervals were descriptive. The protocol followed exploratory results but preceded execution on new seeds. Supplementary Section S5 and Figs. S19–S21 give full rules, development, wider-bound and local-gate controls; the context-cancellation diagnostic is in Fig. 5G.

Serial depth, sensor fidelity and optimization

Physical-depth networks contain 64 somata and eight nonsomatic modules each. D1–D4 reallocate these modules across branching vectors [8], [2, 3], [2, 1, 2] and [1, 1, 2, 2] in the three- and four-tier comparisons; the two-tier D2 tree uses [4, 1]. Contacts and resources are matched; serial composition changes. Grouped-point controls connect every module directly to the soma. Raw-additive controls retain the graph without division, while flexible MLP controls are approximately parameter matched.

Excitatory blocks carry class evidence multiplied by lognormal nuisance factors. Inhibitory blocks sense those factors with sensor fidelity α : zero removes trial-specific information while preserving the sensor distribution, and one reports the factor. The generator floors noisy input rates before multiplying them by gains. Nested, flat and local-ratio families distinguish distributed hierarchical support from generic factor cancellation and locally available ratios. Full generators, floors, sensor-placement controls and contact inventories are defined in Supplementary Section S6, and the calibration fixing this study’s operating point is in Supplementary Fig. S23.

Original cohorts use ten paired seeds, a 180-epoch cap, thirty-epoch patience and validation-selected states. BP and LocalCA differ in rates and parameter groups; exact-path LocalCA still computes exact selected gradients. Source checks of one hundred original trajectories verified configurations and final metrics: all fifty D3 fits across five credit/optimizer arms reached the cap with falling validation loss. Sixty source-consistent restarts extend these fifty D3 fits and ten exact-BP D1 references to 600 epochs, selecting the validation-selected state (minimum validation loss) within each budget. All D3 fits and two D1 references reach the larger cap; eight D1 references stop early. D3 selected states lie near the cap while validation loss continues falling. Supplementary Section S6 retains complete histories, paired accuracy/loss contrasts and source-concordance discrepancies.

Anatomical dictionaries and common-mode controls

Observed MICrONS skeletons and mapped contacts define soma-rooted branch-to-branch segments. Initial eight and disjoint 47-cell cohorts come from the same mouse; Pinky is a second mouse, with ten of twelve selected cells meeting the inherited typed-contact criterion. Each subtree route contains the input-bearing descendants of its origin. Contact labels are direct where available and otherwise proxy-based as specified in Supplementary Section S7. Compression preserves path length but only approximates resistance; direct-label, mapping and geometry sensitivities are retained.

The modeled passive perturbation field consists of reciprocal passive-cable gradient changes induced by small focal shunts. Every corrected dictionary receives one constant column and $K - 1$ spatial columns. Ancestry selection uses inhibitory fraction times excitatory-area-weighted descendant fraction without field outcomes. Depth bins are reparameterized with the same constant while preserving their span; random routes, independent site shuffles and degree-depth surrogate trees provide controls. The weighted common component is removed before computing residual-energy capture. A common-constrained SVD oracle supplies the rank-budget ceiling. Numerical rank, coverage and nonzero density are reported separately; equal column count alone does not match those resources.

The initial eight cells provided post hoc development for common-mode correction; the protocol preceded evaluation of new v661 and Pinky outcomes. Two hundred randomized controls are averaged within each cell. At eight columns, all 47 v661 cells and eight Pinky cells qualify; other eligible budgets are retained without substituting for missing eight-column outcomes. Paired cell-bootstrap intervals and Holm-adjusted comparisons describe within-cohort consistency, not animal-population replication. The older route-generated positive controls use separate fields and endpoint definitions (Supplementary Section S7).

Focal shunting and the electrotonic boundary

Up to sixteen focal sites per cell span depth quartiles and require at least three excitatory descendants and three comparison sites. Normalized E/I scales are 0.35 with reversals one and -0.2 . Shunt doses are 0.25, 0.5, one and two times local membrane-plus-synaptic conductance. A matched injection supplies first-order current $\kappa_k(E_1 - V_k)$ without changing the conductance matrix. A separately solved somatic current restores baseline voltage and is held fixed during differentiation. The squared-loss target is one normalized unit below baseline; finite differences check the adjoint gradients.

Localization subtracts depth-matched off-route from descendant medians of $|\log[(|\gamma'_i| + \varepsilon)/(|\gamma_i| + \varepsilon)]|$, with $\varepsilon = 10^{-15} \max(1, \max_i |\gamma_i|)$. Because this is unsigned, signed gradient changes are reported separately. Adjoint and driving-force substitutions in $\gamma_i = q_i(E_E - V_i)$ distinguish transport from local voltage effects. Sites and relation-reassignment templates are nested within cells. The passive ancestry-partition identity and its dictionary qualification are derived in Supplementary Section S8.

Physical calibration uses cylindrical axial conductance $\pi r^2/(R_a \ell)$ and leak $2\pi r \ell/R_m$, expressed in nS, with spherical soma area $4\pi r^2$. Here r and ℓ are segment radius and length, R_a axial resistivity and R_m specific membrane resistance. Sweeps include $R_a = 100, 150, 250 \Omega \text{cm}$ and $R_m = 300\text{--}30,000 \Omega \text{cm}^2$, fixed and locally normalized doses, and background conductance. The disjoint calibration cohort contains 235 eligible sites in 45 cells. Geometric fallbacks and dose definitions are specified in Supplementary Section S8.

The weak-channel check uses $R_a = 150$, $R_m = 1,000$ in those units and background conductance equal to leak. Median maximal Na, K, Ca and HCN conductances are 0.06, 0.10, 0.025 and 0.05 times leak; NMDA is 0.15 times excitatory conductance. Sixty-four draws per cell passed the predefined steady-state checks. Responses use rank-one updates of the fixed baseline Jacobian, including voltage-gate derivatives, followed by somatic state matching. They are local linearizations rather than newly solved nonlinear equilibria at each dose (Supplementary Fig. S30D,E).

Measured-response prediction and geometric diagnostics

The seven-target cohort includes thirteen eligible scans. Trial responses are the mean of the supplied ROI fluorescence traces in `RoiResponseSeries` over each stimulus interval; our analysis did not deconvolve the traces or apply a reliability correction. Repeated responses are averaged by stimulus condition for the ancestry-similarity analysis; partial rank correlations control Euclidean separation and soma-to-contact path-length differences. Offline learning holds out stimulus identities. Training-only percentile transforms produce nonnegative presynaptic proxies, and each partner enters at its dominant mapped contact segment. A linear somatic readout predicts the standardized target response. Only input-conductance credit differs across rules; readout and bias gradients remain exact.

Four-route dictionaries are selected without target responses and receive the projected baseline transfer profile multiplied by the scalar readout error. Supplementary Section S9 defines delivery and reconstruction. Fig. S31 reconstructs supports from saved segment identifiers, illustrating the median-input-count scan with identifier-based tie breaking; all scan supports are available. At common exact-trained states, field and eligibility-weighted update reconstruction compare the fixed profile, per-trial field projection and the best within-span update. These are oracle diagnostics rather than newly trained rules. Reproducibility comparisons report endpoint discrepancies across 130 exact-rule re-executions.

Ridge regularization uses training data within each outer split and the same measured predictors. Input removal, predictor noise and reliability masks are separate sensitivities; none uses held-out information for selection. Ten split/initialization replicates are averaged within scans and then within seven targets. Imposed route-span-alignment quadratic fields and the six-animal external signed-credit reanalysis are separate datasets and tests, described in Supplementary Sections S7 and S9 and Supplementary Fig. S33, respectively.

Statistics and computational provenance

Confirmatory choices precede outcome inspection but are not described as public preregistration. Pilots, post hoc analyses and same-seed reruns are identified separately. Effect sizes accompany experiment-specific 95% intervals, or adjusted 97.5% intervals where prespecified. Tests are two sided unless a directional protocol states otherwise. Named multiplicity families and exact tie treatment are retained in Source Data; null results imply equivalence only when a margin is tested.

For the sensitivity simulation, a positive-semidefinite shared-path covariance generated joint tuning across the 102 observed presynaptic roots. We mixed ancestry-correlated and independent reliable tuning, then added noise calibrated to each recording’s split-half reliability and actual repeat counts. Four thousand paired datasets at each of 21 signal fractions were analyzed with the original partial-rank statistic and an exact seven-target signed-rank test. Detection required two-sided $P \leq 0.05$ and a positive mean effect; Wilson intervals describe Monte Carlo uncertainty. The model and analysis were frozen before simulated outcomes. Supplementary Section S9 gives the covariance, reliability calibration, dependence assumptions and exact test resolution; the released pair-level inputs permit reproduction.

Independent units are paired training seeds, reconstructed cells, postsynaptic targets or external animals. Sites, contacts, scans, splits, checkpoints and randomized draws remain nested. Seed counts reflect the experiment’s purpose; no prospective parametric power calculation was performed. Source hashes certify retained inputs and implementations, not unavailable historical runtime states. The matched credit and new conductance studies recorded Python 3.10.13, NumPy 2.2.6 and pandas 2.3.3. The new MNIST controls used PyTorch 2.9.1/CUDA 12.8 and torchvision 0.24.1 with NumPy 2.2.6; the depth extension used Python 3.10 and PyTorch 2.9.1. The isolated reviewer environment was validated with NumPy 1.26.4, SciPy 1.15.3, PyTorch 2.9.1, torchvision 0.24.1 and Matplotlib 3.10.9. Cohort-specific runtime records distinguish executed experiments from this installation check. Supplementary Section S10 consolidates statistical families and release records; the question-specific sections retain their exclusions and numerical checks.

OpenAI Codex assisted with code, statistical and consistency audits, figure preparation, literature checks and language editing. The authors remain responsible for design, interpretation, numerical accuracy, citations and the final manuscript.

Ethics and data reuse

No new experiments involving animals or human participants were performed. All biological analyses reused public MICrONS, DANDI, and published animal source data. The source studies describe the original data collection and institutional approvals [22, 42, 63].

Data availability

MICrONS anatomy and connectivity are available through the public CAFE datastack minnie65_public. The primary cohort used materialization 1822, skeleton-service reconstructions, and the synapses_pni_2 table. The disjoint sensitivity cohort used the public v661 static release. Secondary target-structure labels used only in the full-contact analysis came from the in-development synapse_target_predictions_ssa_v2 table. These labels are distributed only as derived data and remain subject to the source-data access terms. All principal structural findings were repeated without them.

The independent-animal replication used the MICrONS phase-1 layer-2/3 Pinky v185 fixed meshes, soma-valence table, and dense synapse table archived at Zenodo DOI 10.5281/zenodo.3710459 [64]. File sizes, Zenodo MD5 checksums, and local SHA-256 checksums are recorded in the supplied manifest. Volumetric imagery is archived in BossDB under DOI 10.60533/BOSS-2021-T0SY, and co-registered functional data are available as DANDI:000402. The supplied manifest records the exact asset and table identifiers, and the source publication describes the full data release [42].

Public benchmark datasets are available from their original repositories [60–62]. The inhibitory-census tables and SWCs are available with the source record for Schneider-Mizell et al. [63]. The Francioni et al. workbook is available through the Source Data link associated with the Nature article [22]; its checksum and the sheet names used here are recorded in the manifest. Derived cell and exclusion manifests, numerical source data underlying every figure, and non-restricted summaries are supplied with this manuscript. The measured-response release includes the analyzed partner-pair tables, contact geometry, recording identifiers and repeated-stimulus response arrays needed for the sensitivity analysis. Authentication tokens and externally hosted raw caches are not redistributed.

Code availability

The accompanying reviewer software archive contains the complete training implementation, exact-gradient and perturbation analyses, versioned configurations, tests, figure scripts, and byte-identified snapshots of the critical training components under the MIT License. A separate Source Data archive contains the numerical data underlying every display item. These archives are available to editors and reviewers with the submission. The same version will be released publicly under the MIT License in a versioned repository and permanent archival record upon publication.

References

- [1] Wilfrid Rall. Theory of physiological properties of dendrites. *Annals of the New York Academy of Sciences*, 96(4): 1071–1092, 1962. doi: 10.1111/j.1749-6632.1962.tb54120.x.
- [2] Christof Koch. *Biophysics of Computation: Information Processing in Single Neurons*. Oxford University Press, 1999. doi: 10.1093/oso/9780195104912.001.0001.
- [3] Panayiota Poirazi, Terrence Brannon, and Bartlett W. Mel. Pyramidal neuron as two-layer neural network. *Neuron*, 37(6):989–999, 2003. doi: 10.1016/S0896-6273(03)00149-1.
- [4] Michael London and Michael Häusser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005. doi: 10.1146/annurev.neuro.28.061604.135703.
- [5] Nelson Spruston. Pyramidal neurons: dendritic structure and synaptic integration. *Nature Reviews Neuroscience*, 9: 206–221, 2008. doi: 10.1038/nrn2286.
- [6] Tiago Branco and Michael Häusser. The single dendritic branch as a fundamental functional unit in the nervous system. *Current Opinion in Neurobiology*, 20(4):494–502, 2010. doi: 10.1016/j.conb.2010.07.009.
- [7] Matthew Larkum. A cellular mechanism for cortical associations: an organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3):141–151, 2013. doi: 10.1016/j.tins.2012.11.006.
- [8] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. Learning representations by back-propagating errors. *Nature*, 323:533–536, 1986. doi: 10.1038/323533a0.
- [9] Luis B. Almeida. A learning rule for asynchronous perceptrons with feedback in a combinatorial environment. In *Proceedings of the First IEEE International Conference on Neural Networks*, volume 2, pages 609–618, 1987.
- [10] Fernando J. Pineda. Generalization of back-propagation to recurrent neural networks. *Physical Review Letters*, 59 (19):2229–2232, 1987. doi: 10.1103/PhysRevLett.59.2229.
- [11] Donald O. Hebb. *The Organization of Behavior: A Neuropsychological Theory*. Wiley, New York, 1949.
- [12] Nicolas Frémaux and Wulfram Gerstner. Neuromodulated spike-timing-dependent plasticity, and theory of three-factor learning rules. *Frontiers in Neural Circuits*, 9:85, 2016. doi: 10.3389/fncir.2015.00085.
- [13] Timothy P. Lillicrap, Daniel Cownden, Douglas B. Tweed, and Colin J. Akerman. Random synaptic feedback weights support error backpropagation for deep learning. *Nature Communications*, 7:13276, 2016. doi: 10.1038/ncomms13276.
- [14] Arild Nøkland. Direct feedback alignment provides learning in deep neural networks. In *Advances in Neural Information Processing Systems*, volume 29, pages 1037–1045, 2016.
- [15] Benjamin Scellier and Yoshua Bengio. Equilibrium propagation: bridging the gap between energy-based models and backpropagation. *Frontiers in Computational Neuroscience*, 11:24, 2017. doi: 10.3389/fncom.2017.00024.
- [16] James C. R. Whittington and Rafal Bogacz. An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation*, 29(5):1229–1262, 2017. doi: 10.1162/NECO_a_00949.
- [17] Yuhang Song, Beren Millidge, Tommaso Salvatori, Thomas Lukasiewicz, Zhenghua Xu, and Rafal Bogacz. Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27: 348–358, 2024. doi: 10.1038/s41593-023-01514-1.
- [18] Guillaume Bellec, Franz Scherr, Anand Subramoney, Elias Hajek, Darjan Salaj, Robert Legenstein, and Wolfgang Maass. A solution to the learning dilemma for recurrent networks of spiking neurons. *Nature Communications*, 11: 3625, 2020. doi: 10.1038/s41467-020-17236-y.

- [19] Justin Werfel, Xiaohui Xie, and H. Sebastian Seung. Learning curves for stochastic gradient descent in linear feedforward networks. *Neural Computation*, 17(12):2699–2718, 2005. doi: 10.1162/089976605774320539.
- [20] James C. R. Whittington and Rafal Bogacz. Theories of error back-propagation in the brain. *Trends in Cognitive Sciences*, 23(3):235–250, 2019. doi: 10.1016/j.tics.2018.12.005.
- [21] Timothy P. Lillicrap, Adam Santoro, Luke Marris, Colin J. Akerman, and Geoffrey Hinton. Backpropagation and the brain. *Nature Reviews Neuroscience*, 21:335–346, 2020. doi: 10.1038/s41583-020-0277-3.
- [22] Valerio Francioni, Vincent D. Tang, Enrique H. S. Toloza, Zilan Ding, Norma J. Brown, and Mark T. Harnett. Vectorized instructive signals in cortical dendrites. *Nature*, 652:1254–1263, 2026. doi: 10.1038/s41586-026-10190-7.
- [23] C. Koch, T. Poggio, and V. Torre. Nonlinear interactions in a dendritic tree: localization, timing, and role in information processing. *Proceedings of the National Academy of Sciences USA*, 80(9):2799–2802, 1983. doi: 10.1073/pnas.80.9.2799.
- [24] Gary R. Holt and Christof Koch. Shunting inhibition does not have a divisive effect on firing rates. *Neural Computation*, 9(5):1001–1013, 1997. doi: 10.1162/neco.1997.9.5.1001.
- [25] Frances S. Chance, L. F. Abbott, and Alex D. Reyes. Gain modulation from background synaptic input. *Neuron*, 35(4):773–782, 2002. doi: 10.1016/S0896-6273(02)00820-6.
- [26] Albert Gidon and Idan Segev. Principles governing the operation of synaptic inhibition in dendrites. *Neuron*, 75(2):330–341, 2012. doi: 10.1016/j.neuron.2012.05.015.
- [27] Matthew Lovett-Barron, Gergely F. Turi, Patrick Kaifosh, Peter H. Lee, Frédéric Bolze, Xiao-Hua Sun, Jean-François Nicoud, Boris V. Zemelman, Scott M. Sternson, and Attila Losonczy. Regulation of neuronal input transformations by tunable dendritic inhibition. *Nature Neuroscience*, 15:423–430, 2012. doi: 10.1038/nn.3024.
- [28] Panayiota Poirazi and Bartlett W. Mel. Impact of active dendrites and structural plasticity on the memory capacity of neural tissue. *Neuron*, 29(3):779–796, 2001. doi: 10.1016/S0896-6273(01)00252-5.
- [29] Robert Legenstein and Wolfgang Maass. Branch-specific plasticity enables self-organization of nonlinear computation in single neurons. *Journal of Neuroscience*, 31(30):10787–10802, 2011. doi: 10.1523/JNEUROSCI.5684-10.2011.
- [30] Ilenna Simone Jones and Konrad Paul Kording. Might a single neuron solve interesting machine learning problems through successive computations on its dendritic tree? *Neural Computation*, 33(6):1554–1571, 2021. doi: 10.1162/neco_a_01390.
- [31] David Beniaguev, Idan Segev, and Michael London. Single cortical neurons as deep artificial neural networks. *Neuron*, 109(17):2727–2739.e3, 2021. doi: 10.1016/j.neuron.2021.07.002.
- [32] Robert Urbanczik and Walter Senn. Learning by the dendritic prediction of somatic spiking. *Neuron*, 81(3):521–528, 2014. doi: 10.1016/j.neuron.2013.11.030.
- [33] Mathieu Schiess, Robert Urbanczik, and Walter Senn. Somato-dendritic synaptic plasticity and error-backpropagation in active dendrites. *PLOS Computational Biology*, 12(2):e1004638, 2016. doi: 10.1371/journal.pcbi.1004638.
- [34] Jordan Guerguiev, Timothy P. Lillicrap, and Blake A. Richards. Towards deep learning with segregated dendrites. *eLife*, 6:e22901, 2017. doi: 10.7554/eLife.22901.
- [35] João Sacramento, Rui Ponte Costa, Yoshua Bengio, and Walter Senn. Dendritic cortical microcircuits approximate the backpropagation algorithm. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- [36] Alexandre Payeur, Jordan Guerguiev, Friedemann Zenke, Blake A. Richards, and Richard Naud. Burst-dependent synaptic plasticity can coordinate learning in hierarchical circuits. *Nature Neuroscience*, 24:1010–1019, 2021. doi: 10.1038/s41593-021-00857-x.
- [37] Benjamin Ellenberger, Paul Haider, Federico Benitez, Jakob Jordan, Kevin Max, Ismael Jaras, Laura Kriener, and Mihai A. Petrovici. Backpropagation through space, time and the brain. *Nature Communications*, 17:66, 2026. doi: 10.1038/s41467-025-66666-z.
- [38] Blake A. Richards and Timothy P. Lillicrap. Dendritic solutions to the credit assignment problem. *Current Opinion in Neurobiology*, 54:28–36, 2019. doi: 10.1016/j.conb.2018.08.003.
- [39] Brendan A. Bicknell and Michael Häusser. A synaptic learning rule for exploiting nonlinear dendritic computation.

916 *Neuron*, 109(24):4001–4017.e10, 2021. doi: 10.1016/j.neuron.2021.09.044.

917 [40] Houman Safaai, Maceo Richards, and Bernardo L. Sabatini. Shunting inhibition and dendritic branching shape
918 local credit assignment. *arXiv preprint arXiv:2607.03556*, 2026. URL <https://arxiv.org/abs/2607.03556>.

919 [41] Matteo Carandini and David J. Heeger. Normalization as a canonical neural computation. *Nature Reviews*
920 *Neuroscience*, 13:51–62, 2012. doi: 10.1038/nrn3136.

921 [42] MICrONS Consortium. Functional connectomics spanning multiple areas of mouse visual cortex. *Nature*, 640:
922 435–447, 2025. doi: 10.1038/s41586-025-08790-w.

923 [43] Alain Destexhe, Michael Rudolph, and Denis Paré. The high-conductance state of neocortical neurons in vivo.
924 *Nature Reviews Neuroscience*, 4:739–751, 2003. doi: 10.1038/nrn1198.

925 [44] Zhuokun Ding, Paul G. Fahey, Stelios Papadopoulos, Eric Y. Wang, Brendan Celii, Christos Papadopoulos,
926 Andersen Chang, Alexander B. Kunin, et al. Functional connectomics reveals general wiring rule in mouse visual
927 cortex. *Nature*, 640:459–469, 2025. doi: 10.1038/s41586-025-08840-3.

928 [45] Wolfgang Hackbusch and Stefan Kühn. A new scheme for the tensor representation. *Journal of Fourier Analysis*
929 *and Applications*, 15:706–722, 2009. doi: 10.1007/s00041-009-9094-9.

930 [46] Lars Grasedyck. Hierarchical singular value decomposition of tensors. *SIAM Journal on Matrix Analysis and*
931 *Applications*, 31(4):2029–2054, 2010. doi: 10.1137/090764189.

932 [47] Nadav Cohen, Or Sharir, and Amnon Shashua. On the expressive power of deep learning: A tensor analysis. In
933 *Proceedings of the 29th Conference on Learning Theory*, volume 49 of *Proceedings of Machine Learning Research*,
934 pages 698–728, 2016. URL <https://proceedings.mlr.press/v49/cohen16.html>.

935 [48] Julian Rossbroich and Friedemann Zenke. Dis-inhibitory neuronal circuits can control the sign of synaptic plasticity.
936 In *Advances in Neural Information Processing Systems*, volume 36, pages 64059–64082. Curran Associates, Inc.,
937 2023. doi: 10.52202/075280-2799.

938 [49] Alessandro R. Galloni, Ajay Peddada, Yash Chennawar, and Aaron D. Milstein. Cellular and subcellular special-
939 ization enables biology-constrained deep learning. *Cell Reports*, 45(4):117159, 2026. doi: 10.1016/j.celrep.2026.
940 117159.

941 [50] Will Greedy, Heng Wei Zhu, Alexia Duriez, Joseph Pemberton, Patrick T. McCarthy, Kevin Kermani Nejad, and
942 Rui Ponte Costa. Cell-type-specific cortical feedback coordinates hierarchical credit assignment. *bioRxiv*, 2026.
943 doi: 10.64898/2026.06.16.732595.

944 [51] Houman Safaai, Maceo Richards, Naeem Khoshnevis, and Bernardo L. Sabatini. When branch-local shunting
945 helps: A gain-load-alignment principle for dendritic E/I networks. *arXiv preprint arXiv:2607.24990*, 2026. URL
946 <https://arxiv.org/abs/2607.24990>.

947 [52] Joseph Cichon and Wen-Biao Gan. Branch-specific dendritic ca2+ spikes cause persistent synaptic plasticity.
948 *Nature*, 520:180–185, 2015. doi: 10.1038/nature14251.

949 [53] Erik B. Bloss, Mark S. Cembrowski, Bill Karsh, Jennifer Colonell, Richard D. Fetter, and Nelson Spruston.
950 Structured dendritic inhibition supports branch-selective integration in CA1 pyramidal cells. *Neuron*, 89(5):
951 1016–1030, 2016. doi: 10.1016/j.neuron.2016.01.029.

952 [54] Spyridon Chavlis and Panayiota Poirazi. Dendrites endow artificial neural networks with accurate, robust and
953 parameter-efficient learning. *Nature Communications*, 16:943, 2025. doi: 10.1038/s41467-025-56297-9.

954 [55] Joel Veness, Tor Lattimore, David Budden, Avishkar Bhoopchand, Christopher Mattern, Agnieszka Grabska-
955 Barwinska, Eren Sezener, Jianan Wang, Peter Toth, Simon Schmitt, and Marcus Hutter. Gated linear networks.
956 In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 10015–10023, 2021. doi:
957 10.1609/aaai.v35i11.17202.

958 [56] Abhiram Iyer, Karan Grewal, Akash Velu, Lucas Oliveira Souza, Jeremy Forest, and Subutai Ahmad. Avoiding
959 catastrophe: active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurorobotics*, 16:
960 846219, 2022. doi: 10.3389/fnbot.2022.846219.

961 [57] Changze Lv, Jingwen Xu, Yiyang Lu, Xiaohua Wang, Zhenghua Wang, Zhibo Xu, Di Yu, Xin Du, Xiaoqing Zheng,
962 and Xuanjing Huang. Dendritic localized learning: Toward biologically plausible algorithm. In *Proceedings of the*
963 *42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*,
964 pages 41682–41700. PMLR, 2025. URL <https://proceedings.mlr.press/v267/lv25c.html>.

- [58] Ken C. Bittner, Aaron D. Milstein, Christine Grienberger, Sandro Romani, and Jeffrey C. Magee. Behavioral time scale synaptic plasticity underlies CA1 place fields. *Science*, 357(6355):1033–1036, 2017. doi: 10.1126/science.aan3846.
- [59] Jeffrey C. Magee and Christine Grienberger. Synaptic plasticity forms and functions. *Annual Review of Neuroscience*, 43:95–117, 2020. doi: 10.1146/annurev-neuro-090919-022842.
- [60] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791.
- [61] Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. arXiv preprint arXiv:1708.07747, 2017.
- [62] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.
- [63] Casey M. Schneider-Mizell, Agnes L. Bodor, Derrick Brittain, JoAnn Buchanan, Daniel J. Bumbarger, Leila Elabbady, Clare Gamlin, Daniel Kapner, Sam Kinn, Gayathri Mahalingam, Sharmishta Seshamani, Shelby Suckow, Marc Takeno, Russel Torres, Wenjing Yin, Sven Dorkenwald, J. Alexander Bae, Manuel A. Castro, Akhilesh Halageri, Zhen Jia, Chris Jordan, Nico Kemnitz, Kisuk Lee, Kai Li, Ran Lu, Thomas Macrina, Eric Mitchell, Shanka Subhra Mondal, Shang Mu, Barak Nehoran, Sergiy Popovych, William Silversmith, Nicholas L. Turner, William Wong, Jingpeng Wu, Jacob Reimer, Andreas S. Tolia, H. Sebastian Seung, R. Clay Reid, Forrest Collman, and Nuno Maçarico da Costa. Inhibitory specificity from a connectomic census of mouse visual cortex. *Nature*, 640:448–458, 2025. doi: 10.1038/s41586-024-07780-8.
- [64] MICrONS Consortium. MICrONS Layer 2/3 Data Tables, 2020. URL <https://doi.org/10.5281/zenodo.3710459>.

Acknowledgements

We thank the MICrONS Consortium and the maintainers of the associated CAVE and DANDI resources. We also thank the authors of the published animal-learning studies for making their source data available. This work was supported in part by a gift from the Chan Zuckerberg Initiative Foundation to establish the Kempner Institute for the Study of Natural and Artificial Intelligence at Harvard University.

Author contributions

H.S. developed the computational framework, performed the analyses, and wrote the initial manuscript. M.R. contributed to the theory of local credit assignment, its implementation, and its interpretation. B.L.S. supervised the work and contributed to the biological framing, study design, and manuscript. All authors reviewed and revised the manuscript.

Competing interests

The authors declare no competing interests.